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Article in Recherche et Applications en Marketing (English Edition) · April 2025

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Quantifying relationships between ordinal categorical variables. Application to metrics tracked by satisfaction barometers.

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Forthcoming in: *Recherche et Applications en Marketing*

Abstract

Perceptions, attitudes, intentions, etc. are routinely studied for market segmentation, tracking and predicting consumer reactions. These cognitive, affective or behavioral components are most often measured by responses to multiple-choice questions: modalities that take the form of a limited number of ordered classes with anchor points, but not always all semantically differentiated. Statistical analysis of such qualitative, ordinal categorical data usually leads us to code them numerically and consider them quantitative. In doing so, we endow them with properties that they do not possess, which entails a risk of bias. To handle them correctly, we propose an econometric parametrization of their relationships, strictly respecting their information content.

Combining cumulative and monotonic multinomial regression with random resampling techniques, our methodology allows us to rigorously test whether ordinal categorical feedback metrics can be assimilated to interval measures. We implement this approach conclusively on two large samples derived from the marketing barometers of two competing telecom operators. Thereby, we show to what extent their customers' willingness to recommend them is explained by their satisfaction with the products and services these two players offer. As a byproduct, this double case study invalidates the very popular *Net Promoter Score (NPS)*, which is supposed to synthesize distributions of degrees to which consumers are likely to participate in the promotion of their favorite brands by word-of-mouth.

Keywords: Ordinal categorical variables. Cumulative and monotone multinomial regression. Random resampling (bootstrapping). Satisfaction barometer. Net Promoter Score (NPS).

Acknowledgements

We are deeply indebted to Professor Laurent BERTRANDIAS, Editor-in-Chief of *Recherche et Applications en Marketing*, who was instrumental in managing the review process of our manuscript. His constructive suggestions, as well as the detailed and thorough comments of the three reviewers he consulted, enabled us to enrich and clarify our contribution.

Introduction

Operationalizing consumer behavior models - linking satisfaction motives (*predictors*) to a manifestation of inclination for a brand (*response*) - requires quantifying these variables, all of which are *qualitative* in nature. This inevitably raises the problem of their numerical coding, whatever the gradation of the scales to which respondents must refer to express their feelings. For example, the intention to repurchase a brand, expressed as a number between 0, for *no chance*, and 100, for *certainly*, could only be treated as a measure of *probability* if

- (1) the perceived gaps between, for example, 0 and 10, 10 and 20, ..., 90 and 100, appear to be equal, so that we can consider these to be quantified according to an *interval* scale,
- (2) and a score twice (or three times, ...) that of another means that customers are twice (or three times, ...) as likely to remain loyal to the brand, so that repurchase intent ratings can be assumed to be calibrated according to a *ratio* scale.

Caution dictates an ordinal interpretation: a higher score simply implies a higher probability. If the proposed scale limits the possibilities of expression to a finite number of categories - for example, from 1 (*not probable*) to 7 (*very probable*) for the likelihood of visiting an online sales site (Mead et al, 2020: 288) - the *responses* must be regarded as *ordinal categorical* (Agresti, 2010) or *ordered categorical* (Anderson and Philips, 1981). It remains to be seen at what cost such *metrics*¹, by their very nature qualitative, can be assimilated to quantitative variables.

Defining “*measurement*” as “*as the assignment of numerals to objects or events according to rules.*”, Stevens (1946: 677) proposed the taxonomy² of “*measurement scales*”. This hierarchy classifies as quantitative data on which calculations of differences (*intervals*), or even *ratios*, make sense and are therefore “*permitted*”. And as *qualitative*, *nominal* (numbers = symbols differentiating the classes of a score) and *ordinal* (numbers = ranks) ones. Most social science research treatises rely on this classification to guide the choice of analysis methods, and warn against improper statistical crushing, particularly of ordinal data³.

¹ Mathematically improper term (reserved for the concept of *distance*) but used as a synonym for quantitative (as opposed to *nonparametric*), widely used to designate marketing indicators (often qualitative), notably by de Haan et al (2015).

² Controversial, but “*became canonical in psychometrics, and from there in statistics for the social sciences more broadly*” (Thomas, 2019: 79).

³ “*illegal*’ statisticizing ...*Outlawing*” (Stevens, op. cit.: 679).

Contextualizing the informational value of data

Like Esteban-Bravo and Vidal-Sanz (2021: 582-585), we consider bans on mathematical manipulations of qualitative data to be too dogmatic, especially when proscribed in the name of Stevens, who was more nuanced than he has been caricatured. We are not condemning practices that have become widespread and which, while lacking in rigor, may prove enlightening, if only for exploratory purposes. Instead, we wish to draw attention to the risks of overestimating the information content of data, which is not determined solely by the meaning of the numbers assigned to observations⁴, but depends just as much on the research objective. For example, Bultez and Derbaix (1982) demonstrated formally and illustrated empirically that for a compensatory linear model, à la Fishbein, to predict a unique ranking in order of preference⁵, the importance of attributes should be measured on a ratio scale, and brands should be evaluated on an interval scale, common to all attributes (up to a constant). This relativization in relation to the object of study does not prevent us from contesting that a score reflecting a degree of agreement or disagreement with a Likert item⁶, be considered an *interval* measurement (Thomas, 2019: 90-91). Paradoxically, it's just as important to accept that undeniably quantitative data must, in some cases, be treated as *ordinal categorical*. This is the case with delivery times (in days or hours), whether we're interested in how customers feel about them (we'll then also measure their perception) or whether we want to assess the logistical efforts required to reduce them. Indeed, it's easier to shorten lead times by one day when they usually last a week than when they rarely exceed forty-eight hours. These should therefore be treated as ordinal categorical measures. Winkelmann (2008: 68), a renowned econometrician, doesn't cite any examples, but considers it favorably: “*Ordered models in general provide a better fit to the data [i.e., counts] than pure count data models [Poisson regression].*” Bultez and Seghers (2025) report a case study supporting Winkelmann's view.

⁴ “*What one ‘finds out’ from one’s data is a function of two things: the information in the data and how this information is extracted.*” (Coombs, 1953: 471).

⁵ Invariant up to licit transformations of its components.

⁶ Like the one devised to measure trust in digital word-of-mouth (Weitzl, 2017: 180-183).

Purpose of our contribution

Rather than blaming those who treat ordinal categorical data as quantitative data, we develop a methodology that allows them to question their practice. Specifically, we

(1) Model the relationships

- between an ordinal categorical response, such as the intention to become a brand ambassador to third parties,
- and its multiple predictors, also ordinal categorical, such as satisfaction with brand attributes,

while all strictly respect the informative value of their measures.

(2) Econometrically calibrate these relationships.

(3) Evaluate whether predictors or responses cannot be treated as quantitative variables; secondly, our approach also demonstrates how to empirically test the validity of the *Net Promoter Score (NPS)*.

Plan

The literature reviewed in the first part of this article justifies the objective we have set. Our approach is described in the second part. The *cumulative multinomial regression*, designed to explain an ordinal categorical *response* by quantitative or nominal variables, is generalized to the estimation of the effects of *ordinal categorical predictors*. Statistically optimal, since it maximizes the likelihood of the data, and robust, since it relies on random resampling, our methodology makes it possible to ascertain the properties of the qualitative measures involved and thus to challenge, or validate, their assimilation to quantitative variables. The presentation is supplemented by technical and computer programming appendices, as well as a tutorial:

CATORDREG.xls, concretizing each phase of the model construction and parameterization.

In the third section, the recommended econometric procedures are applied to two large samples representing the customer bases of two operators competing in the telecoms market of a European Union country. This case exploits the half-yearly monitoring of five customer feedback indicators that feed into the strategic dashboard of one of them:

- their satisfaction with the functional performance of their three product-services, and the quality/price ratio of their overall offering;
- and their willingness to recommend their operators to their relations, which is, according to Reichheld (2003: 48), the gage of “*evangelistic loyalty*”, the engine of long-term growth.

The fourth section highlights both the originality and the limitations of our approach.

Tacking stock of the calibration of ordinal categorical variables

Since the popularization of Fishbein's work by Bass and Talarzyk (1972), empirically explaining attitudes and behavioral intentions towards brands has become, and remains, a major concern. Many variants of the original model, multi-attribute, compensatory and linear, have been tested, often in an unorthodox way (Bultez and Derbaix, op. cit.). The development by Fornell (1992) of the Swedish satisfaction barometer and of its American (*ACSI*, *American Customer Satisfaction Index*: Fornell et al, 1996) and European (*ECSI*) spinoffs inaugurated another research tradition on more complete chaining between satisfaction and attitude. This second school builds upon systems of structural, linear equations with latent variables, integrating antecedents (including perceptions of the usefulness of the product or its attributes) and subsequent outcomes (including actual behavior). Table 1 itemizes ten contributions, some of which relate to the first and the others to the second of these streams. Archetypal of the wide variety of empirical studies of the relationships between marketing metrics, they are symptomatic of the issues related to their quantification. This illustrative range of the most common calibration practices results from a systematic exploration of five of the largest scientific journals in our discipline and three others that have published articles or communications relevant to positioning our contribution. Notwithstanding that it covers a long period, from Bettman (1974) to Brusco (2022), there has been little progress in the econometrics. While Bettman already applied *binomial regression* - in its *probit* variant - to explain the probability that a consumer would be satisfied (or not) with the attributes of toothpaste brands, Brusco uses it - in its logit version - to select the essential determinants of the probability of customers' commitment in word-of-mouth about the restaurants they patronize. But not Huang and Sudhir (2021: 326-327) who, for the analysis of attrition, although binary, stick to a linear specification.

Obliviousness of the actual information content of predictive metrics

Dominant trend. While Bettman and Brusco rigorously treat the nominal (binary) response they study, both confuse predictors with quantitative factors, whereas they are ordinal categorical: ratings of the relevant dimensions of brands on 7-point scales. Between these two props, many others do the same with predictors, whether measured by scales from 1 to 4 (Kamakura et al, 2002: for recommendation likelihood), 1 to 5 (Huang and Sudhir, 2021: satisfaction), 1 to 7 (Bayol et al, 2000: overall contentment and meeting expectations; de Haan et al, 2015: satisfaction; Catapano et al, 2022: attribute weights), or from 0 to 10 (de Haan et al, 2015: willingness to recommend. Larivière et al (2016) rely on the 1-to-100 *ACSI*.

Chakraborty et al (2022) rigorously process the ordinal categorical ratings of restaurants on their main attributes.

Table 1. State-of-the-art articles

Authors	Bettman (1974)	Bayol et al (2000)
Objective	Explain the satisfaction provided by each brand for each attribute, according to Fishbein's model.	Building the <i>European Consumer Satisfaction Index (ECSI)</i> , derived from the Swedish index designed by Fornell (1992).
Response(s)	Binary, satisfaction generated: not satisfactory = 0 / satisfactory=1.	Repurchase and recommendation likelihoods (observable components of the latent construct: loyalty), on a scale from 1 (very negative) to 10 (very positive).
Predictors	Bipolar 7-point scales, arbitrarily quantified from -3 to +3, to assess the contributions of attributes and the probabilities that each brand possesses them.	Measures of overall contentment and fulfillment of expectations (observable components of the latent construct: satisfaction), with a scale from 1 (very negative) to 10 (very positive).
Sample	Per attribute: 847 observations = 121 students × 7 brands Category: toothpaste.	250 cell phone users living in an unspecified European country.
Method	Binomial <i>probit</i> regression with an underlying hypothetical concept: “ <i>belief</i> ” that the brand satisfies their needs in relation to this attribute.	Partial least squares applied to linear structural equations explaining causal links between latent variables, themselves determined by linear combinations of manifest indicators.
Authors	Kamakura et al (2002)	Schmitt et al (2010-2012)
Objective	Explain the profitability of banking services using a causal chain.	Explain the lifetime value of bank clients.
Response(s)	Customer behavior: latent construct combining three quantitative manifestations of their lifetime value.	Observed loyalty: 1= positive amount in deposits /0= zero amount in deposits.
Predictors	Bank recommendation likelihood: measured on an unspecified 4-level ordered scale.	Degree of satisfaction: from 1 (<i>not at all satisfied</i>) to 5 (<i>very satisfied</i>)
Sample	5,550 customers of more than 500 branches of a Brazilian bank; distribution proportional to branch size	Telephone survey of 561 customers of a major German bank, with secondary data on customer profiles and transactions.
Method	Linear structural equations parameterized using sample likelihood maximization.	Binomial logistic regression of loyalty on satisfaction score.
Authors	de Haan et al (2015)	Larivière et al (2016)
Objective	Determination of the consumer feedback metrics most predictive of brand loyalty	Study of the relationship between satisfaction, intention to remain loyal and shareholder value.
Response(s)	Self-reported loyalty: 1=remained brand consumer / 0=no longer brand consumer.	Repurchase intention, scored from 1 to 10.
Predictors	Degrees of (1) satisfaction, from 1=very dissatisfied to 7=very satisfied; (2) willingness to recommend, from 0=very unlikely to 10=very likely.	<i>American Customer Satisfaction Index (ACSI)</i> , ranging from 1 to 100.

Sample	Two-wave survey (1 st wave: 6,649; 2 nd : 1,308), of customers from 93 companies, covering a broad spectrum of service sector industries (18), in the Netherlands	Combination of (a) time series on the financial results of 137 companies operating in seven sectors in the USA, (b) and 189,069 measurements collected from these companies' customers.
Method	Binomial logistic regression of the probability of remaining loyal on satisfaction or willingness to recommend.	Hierarchical Bayesian linear model, with three levels, the first of which specifies the moderating causal link between customer satisfaction and repurchase intention, considered as mediating the effect of satisfaction on shareholder value.
Authors	Huang et Sudhir (2021)	Brusco (2022)
Objective	Calibrate the relationship between satisfaction and willingness to recommend.	Explain the recommendation of a restaurant by ten or so of its qualities.
Response(s)	Willingness to recommend to a friend (<i>RTF</i>): from 1 to 10=will definitely. Binary nominal attrition indicator.	Nominal binary recommendation indicator: 1=recommends/0=doesn't recommend.
Predictors	Satisfaction scored between 1 and 5.	Customers' agreement with the following items: ease of access, taste of food, atmosphere and service; degrees quantified, using a Likert scale, from 1=strongly disagree to 7=strongly agree.
Sample	Some 42,000 reactions from customers surveyed the day after an interaction with a call service rep of a major credit card issuer.	Not very precise: 1,500 clients.
Method	Two-stage least squares: use of instrumental variables (customer profile and know-how indicators of the rep contacted) to obtain an exogenous measure of satisfaction, itself explaining the <i>RTF</i> and attrition, by linear regression.	Binomial logistic regression.
Authors	Catapano et al (2022)	Chakraborty et al (2022)
Objective	Study the preference for the paper version of a book, rather than its digitized version, based on the relative importance of their attributes.	Use spontaneous (unstructured) reviews and overall ratings from restaurant customers to obtain attribute-by-attribute evaluations.
Response(s)	Ternary encoding of preference: -1 = digital version, 0 = indifference, +1 = paper version.	(1) Utility, specific to each customer who expressed an opinion, underlying his/her evaluations of the restaurants on each attribute. (2) Overall restaurant rating, from 1 to 5 stars.
Predictors	Attribute importance rated on a 7-point scale: from 1 = not at all important, to 7 = extremely important.	Ordinal categorical ratings of restaurants on 5 main attributes, inferred from textual analysis of written comments.
Sample	301 participants recruited via the <i>Amazon Mechanical Turk</i> platform, financially rewarded.	45,652 ratings and substantiating comments, published on the <i>Yelp</i> local business platform, for restaurants rated at least 20 times, by customers who posted at least 5 reviews. Main characteristics of restaurants (price range, average rating obtained in the past, etc.).

Method	Ordinal regressions of the response on each of the predictors, treated as quantitative variables.	Inferences (1) of the latent utilities, for each customer, of the attributes characterizing the restaurants' qualities, based on the cumulative logistic model; (2) the weights attributed by each segment to the attributes, deduced from the linear combination of the restaurants' evaluations of these attributes, à la Fishbein, supposed to determine the given global score, treated as quantitative, notwithstanding its ordinal categorical nature.
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Insufficiently substantiated amalgams.

de Haan et al (2015) stand out by comparing the predictive power of variants of the metrics by which to explain the probability of customer retention: satisfaction (from 1 to 7) and recommendation likelihood (from 0 to 10). First, they treat these scores as numerical. Then, they substitute for them nominal measures, resulting from mergers of response categories:

- a binary indicator variable set to **1** when the subject expresses top satisfaction at levels: 6 or 7, and **0** for any lower score;
- the *Net Promoter Score (NPS)*, individualized by a ternary variable: **−1** for degrees of recommendation likelihood from 0 to 6, **0** for 7 and 8 and **+1** for 9 and 10.

These aggregates are arbitrary and reductive: the second conforms to the segmentation advocated by Reichheld and criticized by Bultez et al (2024). Previously, only Kristensen and Eskildsen (2011) had attempted to empirically verify the *NPS* correctness. They drew on data from a representative Danish insurance customer satisfaction survey, using the *Extended Performance Satisfaction Index* questionnaire, to which they added a question on recommendation likelihood: $RL \in \{0, 10, \dots, 100\}$. They queried the *NPS* based on a vertical bar chart, one per degree of RL , with heights equal to the averages of the self-reported loyalty⁷: SRL (op. cit.: 251, Figure 2). On this mapping – represented in Figure 1, in a perfect square format so that distances are not distorted - they pinpointed three homogeneous clusters, but different from those established by Reichheld. This scatterplot endorses the lower cluster: the main axis of the ellipse enveloping the 5 pairs of lowest RL notes is quasi-horizontal; but disavows the median, which is too heterogeneous: the sub-sample for which $RL = 50$ appears to be related neither to those for which $RL \leq 40$, nor to those for which $RL \in \{60, 70\}$. Lacking information on the sizes of the sub-samples, hence unable to guess the precision of their two indicators, we cannot pursue this counter-analysis.

⁷ In response to the question of whether the respondent will still be a customer of their company in a year's time (from 1=no, absolutely not to 10=yes, absolutely). These scores were recalibrated between 0 and 100.

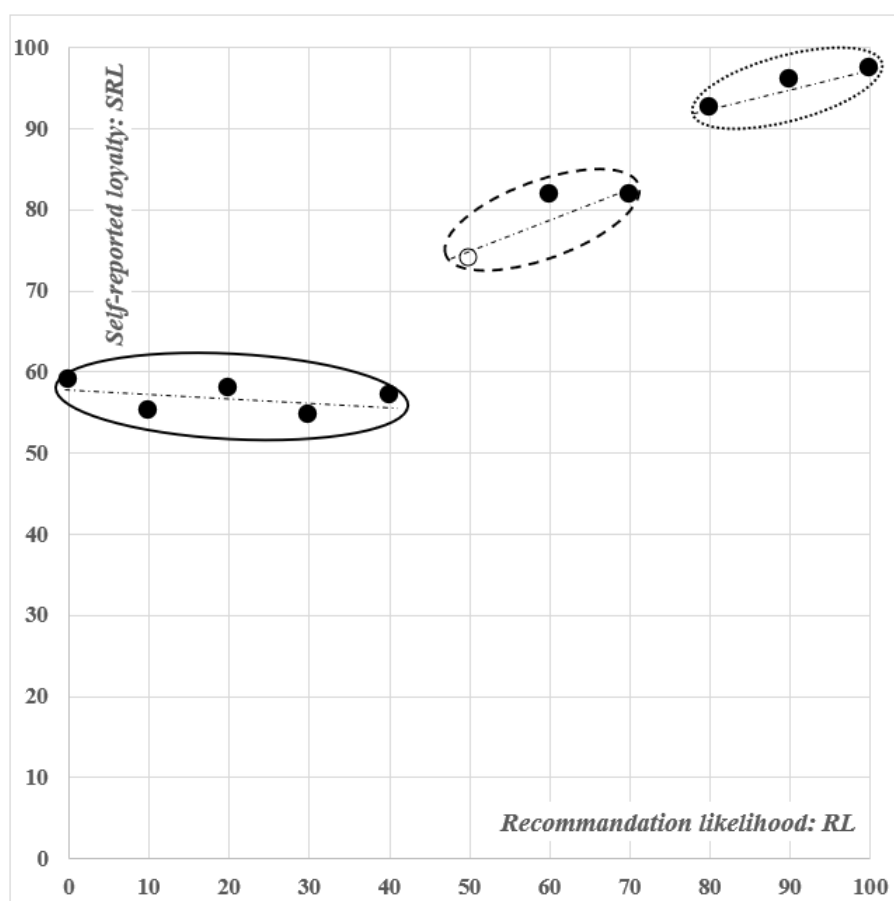


Figure 1. Association between means of *RL* and *SRL*

Grounding their diagnosis on this graphical description of the relationship between two complementary manifestations of loyalty, neither of which can be considered cause of the other, they seem to have proceeded in a similar way to what Reichheld (2003) said he did without documenting it⁸.

Bypassing the ordering. Rather than dichotomizing satisfaction scores into superior (top 2) versus inferior, de Haan et al could have used multiple class-variables to distinguish degrees of their scale. This would have led them to introduce six binary variables: one for each level, except for the one chosen as the reference base then assimilated to the constant term. This is the rigorist route taken by Schmitt et al (2010-2012) to demonstrate that the customers most inclined to recommend their bank⁹ are the most profitable. A study of the coefficients of their four binary auxiliary variables (point estimates and their standard errors presented in the 1st column of their Table 6, op. cit. 138) reveals that

⁸ Initially, Reichheld justified his classification with a single, brief, unfalsifiable contention (2003: 51): “When we examined customer referral and repurchase behaviors along this scale [0-to-10: recommendation likelihood] we found three logical clusters.”

⁹ Predisposition measured by a Likert scale: from 1 = *strongly disagree* with recommending the bank to 5 = *totally agree*.

- the bank actually achieves a lower margin on the transactions with customers least inclined to recommend it (degree of willingness smaller or equal to 3);
- the margins achieved with those most prone to recommend the bank do not appear to be significantly greater than those achieved on the transactions of those who are slightly less inclined (degree 4): the reference level (degree 5) is included in the confidence interval corresponding to degree 4, and the risk of wrongly concluding that these margins for degree 5 would be greater than those for degree 4 is 18.12% (one-tailed p -value), according to our calculations.

Having recourse to binary dummies may lead to aberrations, as the estimates of their coefficients are not constrained to fit the gradation of the ordinal categorical scale used. In the case of Schmitt et al, the order was violated: the coefficient of the lowest degree of willingness (1) was found to be higher than those of degrees 2 and 3. However, without knowledge of the covariances between estimators¹⁰, we cannot judge the seriousness¹¹ of this double anomaly. Schmitt et al (p. 138) analyzed the two extreme variants of the treatment of their scale as nominal: exhaustive specification (without any grouping of levels, which they called: “*standard coding*”) versus exclusive distinction of those most prone to recommend from all others. They concluded that: the “*standard coding of the intention to recommend a company...or the exclusive consideration of customers with the highest intention to recommend (promoters) do not demonstrate a significant influence*”. Had they been concerned about the inconsistencies noted, they would have carried out tests of differences between successive levels, estimating specifications with three binary dummies (grouping the first two levels), then with two (grouping the first three degrees), but after eliminating progressively (backward stepwise) control variables whose effects appeared negligible.

Problematic dichotomization of predictors and response

A number of authoritative scholars, specialists in the analysis of interactions between an experimentally manipulated determinant and a moderating factor considered to be continuous, have strongly condemned the binarization of this third uncontrolled variable by splitting its values into those above and those below its median (Fitzsimons, 2008). Among the most radical opponents to this dichotomization, Rucker et al. (2015: 674) also warn against those who treat the “*measured*” (moderator) and “*dependent*” variables as “*continuous*” when they

¹⁰ Presumably positive between the estimators of the coefficients of the dummies, they reduce the variances of the differences between them.

¹¹ “*The point estimate is generally the effect size best supported by the study, irrespective of its statistical significance/nonsignificance*” (Wasserstein et al, 2019: 14).

are “ordinal (e.g., assessed on a single 11-point scale)” and suggest the use of more advanced methods than ordinary linear regression: “discretizing in a manner that respects the ordinal nature of the data, treating the data as nominal...ordinal regression techniques”. Similarly, McClelland et al (2015: 681) encourage testing for main effects and interactions “via polynomials, other forms of nonlinear regression or latent class models” with “dummy auxiliary” variables.

Bollinger et al (2022) examined the impact of a national campaign sensitizing Canadian consumers to the labelling of food products according to their nutritional value. They tested the effects of this campaign, particularly on consumer *trust* in this grading (op. cit.: 168-169), through a binary logistic regression of *trust* on the nominal indicator distinguishing post-campaign measures from those in the pre-campaign period. To do this, they dichotomized the response levels proposed to express confidence in the labelling: “ ‘On a scale of 1 to 5, where 1 is not trustworthy and 5 is extremely trustworthy, please tell me the extent to which you trust the Guiding Stars symbols on shelf tags for providing food and nutrition information’ [responses of 4 or 5 were categorized as trust].”

Room for improvement in response modeling

Pertinent analysis of dichotomous responses. Bettman (1974), Schmitt et al (2010-2012), de Haan et al (2015) and Brusco (2022) rightly opted for binomial regression.

Assumed cardinality. Responses resulting from a choice between several ordered categories are generally assimilated to quantitative variables of linear progression (Bayol et al., 2000; Larivière et al., 2016; Huang and Sudhir, 2021). This long-decried practice is all the more to be proscribed when its antecedents are studied by ordinary regression. Agresti (2010: 5-7) criticizes this persistent practice¹² because “the ordinary regression approach does not account for ‘ceiling effects’ and ‘floor effects’, which occur because of the upper and lower limits for the ordinal response variable”. He therefore recommends *cumulative multinomial regression*. In marketing, Bultez and Guerra (2005) are, to our knowledge, the first to have applied it, but in an area unrelated to the relationships examined in Table 1: they experimented with the effect of adding a generic to a ready-to-eat food product category on perceptions of dissimilarities between private-label and national brands.

¹² Huang and Sudhir (2021) took the liberty of doing so in a recent article, by the way is very interesting and methodologically sound in every respect. Did they have this weakness in mind when they wrote their last sentence (339): “We did not investigate...whether and how the Likert scale used in measuring satisfaction and RTF [willingness to recommend to a friend] may impact the estimated causal relationship between satisfaction and RTF”?

The complexity of cumulative multinomial regression has hampered its adoption: it is far more sophisticated than binomial logit regression, which it generalizes. The latter - a variant of probit regression (Bettmann, 1974), whose principle is due to Bliss (1934) - although much older, apparently still poses problems for some: Reiff et al. (2023: 1105) admit that they only report the results of conventional linear regressions¹³ "*for ease of interpretation*" rather than those of binomial logistic regressions, which they have also carried out. Not surprisingly, the educational journal edited by *INFORMS* recently published Brusco's (2022) article aimed at facilitating the teaching of the latter basic technique.

Two notable exceptions prove the rule. Catapano et al (2022) studied preferences for physical goods versus their digitized equivalents (books, films, photos, calculators, games). To test the extent to which the mode of elicitation of these preferences - *willingness to pay* versus *choice between the two formats* - could reverse them, they conducted an impressive series of experiments. One of these (pp. 363-364) sought to determine the impact of the weights of five attributes (including *convenience* and *environmental impact*) on the preference for the paper edition of *Harry Potter* over its digital version. To this end, they performed separate *ordinal regressions* of preference on the importance attached to each attribute. Graded from 1 = *not important at all* to 7 = *extremely important*, they treated it as a quantitative predictor, without worrying about its ordinal categorical nature.

Chakraborty et al (2022) remarkably demonstrated how to abstract from an impressive number of feedbacks posted a platform¹⁴, evaluations of restaurants according to their own attributes, by their customers. The model they propose

(a) takes into account the heterogeneity of consumers' expectations and their propensity to focus their comments on a subset of attributes, omitting to mention the others [equation (4): 609] ;

(b) quantifies [equations (2)-(3): 609], on the basis of cumulative logistic regression, the unobservable utilities explaining restaurant evaluations, derived, attribute per attribute, from text-mining reviews, respecting the categorical nature of these evaluations, which can vary from 1 = *extremely negative opinion* to 5 = *extremely positive opinion* [Figure 1: 605];

(c) estimates the importance attached to attributes, by segment (latent class), by postulating a relationship between, on the one hand, global evaluations - ordinal categorical: from 1 to 5 stars - and, on the other hand, the linear combination of restaurant evaluations on attributes [equation (7): 610].

¹³ Contraindicated, because the response they want to explain is binary: 1=the employee has increased his contribution to his retirement savings plan, 0=he hasn't (ibidem: 1104).

¹⁴ <https://fr.yelp.be/biz/le-chef-tha%C3%AF-paris-2?hrid=wtP9qm-k3uD5XsvlhKHnqw&osq=Cuisine+Thai>.

Technically the most advanced of those listed in Table 1, the approach designed and applied on a huge database by Chakraborty et al (2022), deserves the attention their article has attracted, despite its intricacy. It is, however, surprising that they equated the overall rating (c) with a quantitative measure¹⁵, whereas they rigorously treated restaurant ratings on attributes on (b).

Aim of our approach

The mitigated results of our readings prompted us to seek an approach that would allow us to specify and fit correctly any model designed to explain a response of an ordinal categorical nature by antecedents of the same nature. This is why our second part sets out the foundations of cumulative multinomial regression and how it can be generalized, since its standard form does not allow ordinal categorical explanatory factors to be treated as rigorously as the *response* they elicit. This limitation is undoubtedly due to the fact that its champion tolerates equating them with quantitative predictors¹⁶: *“strict adherence to operations that utilize only the ordering in ordinal scales limits the scope of useful methodology [i.e., cumulative multinomial regression] too severely ... Nearly all models assign scores to the categories and regard the variable as quantitative - the alternative being to ignore the ordering and treat the variable as nominal, with indicator variables Therefore, we do not take a rigid view about permissible methodology for ordinal variables.”* (Agresti, 2010: 5). In what follows, we pave the way for another option and, in so doing, explain how we can test whether response and predictors can be considered quantitative variables.

Methodology

The econometric technique developed below aims to rigorously estimate any relationship between ordinal categorical variables, such as those examined in Table 1. However, to avoid excessive abstraction, we focus on the most tested variables: those linking consumer satisfaction with a brand's product/service offering to attitudes toward that brand. The recommended method is designed to use data collected from a sample of N consumers

¹⁵ The quantification of its predictors - step (b) - is not enough to justify such an assimilation.

¹⁶ Following Anderson and Philips (1981: 24): *“in practice, any such variable may be approximated by a continuous variable taking numerical values specified for each category...Alternatively, an ordered categorical variable with q categories may be replaced by $q - 1$ dummy binary variables. This is less attractive for several such variables with many levels as it increases the number of parameters considerably.”*

(identified by an initial index i specific to each individual) who are asked about

- their evaluation of a brand on M relevant criteria (distinguished by a second index m , for the behavioral motive), giving as many expressions of satisfaction: $s_{i,1}, \dots, s_{i,m}, \dots, s_{i,M}$, all with reference to the same ordinal scale with $G + 1$ judgment categories, from least favorable to most favorable, assumed to be numerically coded, without loss of generality:

$$c_0 < c_1 < \dots < c_j < \dots < c_G;$$

- and their preference for the same brand,
 - expressed by reference to another ordinal categorical scale with $D + 1$ judgment classes, from least favorable to most favorable: $0, 1, \dots, r, \dots, D$;
 - considered their response, $\tilde{R}_i$, essentially resulting from their different satisfaction motives and other barely noticeable causes represented by a random component ($\tilde{\epsilon}_i$):

$$\tilde{R}_i(s_{i,1}, s_{i,2}, \dots, s_{i,m}, \dots, s_{i,M}; \tilde{\epsilon}_i), \forall i \in \{1, 2, \dots, N\}. \quad (1)$$

The chosen notation, as it also evokes stimuli, shows that our model is intended to be very general. The parameterization of (1) is based on the principle of data likelihood maximization, according to the specified model, while strictly respecting the ordinal categorical nature of the two kinds of metrics, through optimal quantifications of the grades on their respective scales. The statistical properties of the estimators are then derived empirically through simulations. These allow not only a more accurate assessment of the importance of each predictor, but also the realization of the consequences of assimilating qualitative responses to quantitative measures. Each of the following steps is illustrated in the tutorial: **CATORDREG.xls**, programmed by the first author and accessible online via the *SAGE Journals Online* and *ResearchGate* (DOI: 10.13140/RG.2.2.15927.02721).

Model building

The path taken to specify (1) is shown in Figure 2.

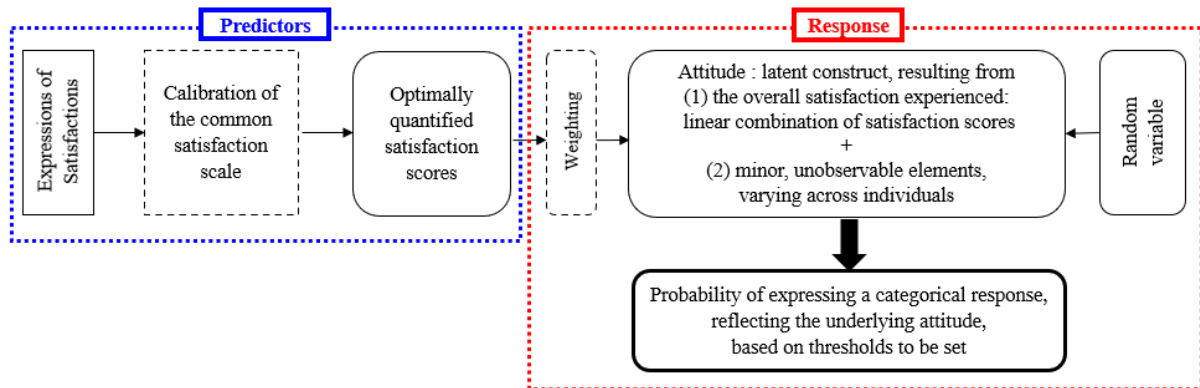


Figure 2. Modeling framework, further formalized in Figure 4.

Predictors. To validly quantify ordinal categorical judgments made about attitude antecedents, the initial c_j must be transformed into quantitatively differentiable measures: κ_j , so that the intervals between these increments can be correctly interpreted. Such conversion first requires a rigorous and flexible recording of the qualitative judgments made by consumers, using *binary variables*:

$$b_{i \rightarrow j|m} = \begin{cases} 1, & \text{si } i \text{ has expressed (symbol: } \rightarrow \text{) a judgment} \\ & \text{at level : } c_j, \text{ on facet: } m; \\ 0, & \text{if his judgment, on the same facet,} \\ & \text{is at another level.} \end{cases}$$

The data coding then leads to the formalization of the expressions of satisfaction by linear combinations of these *indicators*:

$$s_{i,m} = \sum_{j=0}^{j=G} c_j \cdot b_{i \rightarrow j|m}, \forall m \in \{1, 2, \dots, M\} \text{ and } \forall i \in \{1, 2, \dots, N\}, \quad (2)$$

such that if subject i , when evaluating motive m ticks category g , then: $s_{i,m} = c_g$, since:

$b_{i \rightarrow g|m} = 1$ and $b_{i \rightarrow j \neq g|m} = 0$. Then, all that's left is to define the c_j , by positive numbers following their ordering. In most cases, a series of numbers corresponding the numbering of the categories is used:

$$c_j = j, \text{ for } j \in \{0, 1, 2, \dots, G\}. \quad (3)$$

This a priori assignment is arbitrary because it assumes equidistance between the echelons:

$$c_1 - c_0 = \dots = c_j - c_{j-1} = \dots = c_G - c_{G-1} = 1.$$

No doubt, it is more realistic to replace c_j with values to be estimated, respecting the ranking.

Let's call these unknowns K_j , which must satisfy the monotonicity conditions: $K_{j+1} \geq K_j$.

These non-strict inequalities are satisfied if we add a non-negative number: $\Delta_{j|S} \geq 0$, to the value assigned to one category to obtain the value to assign to the next: $K_j = K_{j-1} + \Delta_{j|S}$.

In addition, to ensure the non-negativity of these $\Delta_{j|S}$ -increments, which are likely to be variable, we substitute for each one the square of another addon: $\nabla_{j|S}$, which remains to be determined and can be negative, positive, or zero: $\Delta_{j|S} = \nabla_{j|S}^2$. Finally, to facilitate the

comparison between the initial values $c_j = j$ and the values resulting from their parameterization, their respective anchor points should be aligned by standardizing K_j to κ_j , so that: $\kappa_G = G$, $\kappa_j = K_j \cdot [G/K_G]$ for $1 \leq j \leq G - 1$, and setting the origin to zero:

$K_0 = 0 \Rightarrow \kappa_0 = 0$. The intervals between the so anchored quantified levels are thus related to the non-standardized ones by:

$$\delta_{j|S} = \kappa_j - \kappa_{j-1} = [K_j - K_{j-1}] \cdot [G/K_G] = \Delta_{j|S} \cdot [G/K_G].$$

Table 2 illustrates the quantification procedure described above. Substitution of the parameters κ_j for c_j in (2) specifies how to translate qualitative ratings into scores:

$$s_{i,m} \Rightarrow \mathbb{S}_{i,m} = \sum_{j=0}^{j=G} \kappa_j \cdot b_{i \rightarrow j|m}. \quad (4)$$

Table 2. Monotonic parametrization of a 5-point scale.

Scale		Binary encoding of satisfaction ratings: $b_{i \rightarrow j m}$					Quantification of degrees of satisfaction
		Not satisfied at all	Not quite satisfied	Satisfied	Very satisfied	Extremely satisfied	
Degree of satisfaction	Initial coding: c_j	$b_{i \rightarrow 0 m}$	$b_{i \rightarrow 1 m}$	$b_{i \rightarrow 2 m}$	$b_{i \rightarrow 3 m}$	$b_{i \rightarrow 4 m}$	$\kappa_j = \kappa_{j-1} + \delta_{j S}$ $\delta_{j S} \geq 0$
Not satisfied at all	$c_0 = 0$	1	0	0	0	0	$\kappa_0 = c_0 = 0$
Not quite satisfied	$c_1 = 1$	0	1	0	0	0	$\kappa_1 = \kappa_0 + \delta_{1 S}$
Satisfied	$c_2 = 2$	0	0	1	0	0	$\kappa_2 = \kappa_1 + \delta_{2 S}$
Very satisfied	$c_3 = 3$	0	0	0	1	0	$\kappa_3 = \kappa_2 + \delta_{3 S}$
Extremely satisfied	$c_4 = 4$	0	0	0	0	1	$\kappa_4 = \kappa_3 + \delta_{4 S} = 4$

The shared parametrization of the M evaluation criteria, embodied in equation (4), assumes that consumers are

- constant, i.e. they implicitly refer to one and same scale of values when expressing their level of satisfaction with all features;
- able to subjectively guess the relative magnitudes of the differences between these values¹⁷ without being able to make them explicit.

These hypotheses of minimal rationality in the judgments made must be verified (Bultez and Derbaix, 1982), a fortiori if we are to justify what we usually allow ourselves by assimilating, as in (3), the ordinal categorical scale to an interval one.

Response. The principle of quantitative calibration of predictors, formalized by (4), could also be applied to consumer response: $R_i \Rightarrow \mathbb{R}_i$. We could then imagine estimating the monotonic regression model, of the linear compensatory processing type:

$$\tilde{\mathbb{R}}_i = \varphi_0 + \sum_m \varphi_m \cdot \mathbb{S}_{i,m} + \tilde{\epsilon}_i, \quad (5)$$

where $\tilde{\epsilon}_i$ is the residual uncertainty. But what assumption can be made about the distribution of this random component, given the discrete nature of the variables and their limits of variation? Estimates of φ_m would, in some cases, produce logically unacceptable predictions: continuous (not discrete), below (or above) the scale limits which may also be unattainable.

¹⁷ That is, to evaluate the ratios: $(\delta_{j|S}/\delta_{j-1|S})$.

This lack of robustness forces us to opt for a non-linear relationship that naturally respects the $[0, D]$ -range of the response variation. This is why, following McCullagh (1980: 109-110) and Anderson and Philips (1981: 23), we have chosen the cumulative multinomial model, considering that the response expressed is just the manifestation of an underlying unobservable continuous variable. The distribution of responses, in ordered categories, is then supposed to result from the discretization of this latent continuum, which, in this case, it is natural to call attitude ($\tilde{A}_i$) towards the brand. This purely hypothetical construct results from the overall satisfaction (S_i) of the consumer, whose feeling is subject to random variations ($\tilde{\varepsilon}_i$) inherent to the imperfection of her/his evaluations:

$$\tilde{A}_i = S_i + \tilde{\varepsilon}_i. \quad (6. a)$$

In this equation, the deterministic part is the linear combination of the quantified satisfaction scores ($s_{i,m}$) on the evaluation criteria, whose coefficients ($\omega_m > 0$) are the weights:

$$S_i = \omega_0 + \sum_{m=1}^{m=M} \omega_j \cdot s_{i,m}. \quad (6. b)$$

The probability density function (f) of random disturbances - whose dispersion, denoted by σ , determines the weight of the uncertainty - is assumed to be continuous, symmetric, and with a central tendency equal to zero, so that consumer sensations are unbiased: $E[\tilde{A}_i] = S_i$, and hence: $\tilde{\varepsilon}_i \sim f(0, \sigma)$. Now, the distribution of the disturbance, or more precisely, its distribution function: F , remains to be specified. Initially, econometricians opted for the standardized Gaussian distribution, to which they later preferred the logistic distribution, since the latter, whose shape is very close to the *normal*, has the advantage of being determined analytically (without recourse to numerical integration) by the formula:

$$F(\tilde{\varepsilon}_i = \varepsilon_i) = \Lambda(\varepsilon_i) = 1/[1 + e^{-(\varepsilon_i/\sigma)}]. \quad (6. c)$$

The discretization of the latent construct, $\tilde{A}_i$, is assumed to occur implicitly according to intensity *thresholds*, which delimit categories of possibly unequal amplitudes. Figure 3 displays a hypothetical scattering of these thresholds (τ) along the continuum of values for $\tilde{A}_i$: $-\infty < \tau_1 \leq \dots \leq \tau_r \leq \tau_{r+1} \leq \dots \leq \tau_D < +\infty$.

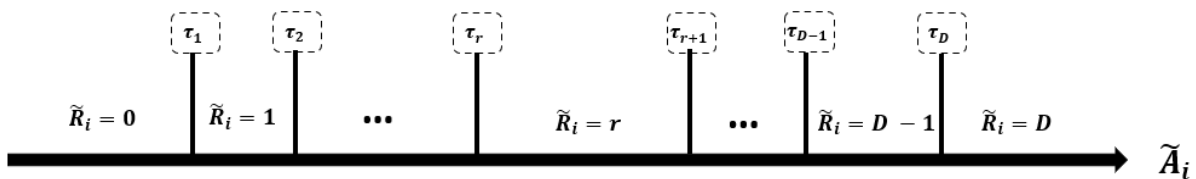


Figure 3. Ordinal categorization of $\tilde{R}_i$

Thus, for a consumer to express a response: r , the strength of his attitude must lie between two limits: τ_r and τ_{r+1} , treated as parameters to be calibrated. Hence, the probability (P) that s/he will express this response is determined, for $r \in \{1, 2, \dots, D-1\}$, by:

$$\pi_{i,r} = P(\tilde{R}_i = r) = P(\tau_r < \tilde{A}_i < \tau_{r+1}) = P(\tilde{A}_i < \tau_{r+1}) - P(\tilde{A}_i < \tau_r). \quad (7.a)$$

The probabilities of extreme responses are just special cases of formula (7.a):

$$\pi_{i,0} = P(\tilde{R}_i = 0) = P(\tilde{A}_i < \tau_0) \quad (7.b) \quad \text{et} \quad \pi_{i,D} = P(\tilde{R}_i = D) = 1 - P(\tilde{A}_i < \tau_D). \quad (7.c)$$

Each of these response probabilities is thus calculated from the cumulative distribution of the values of $\tilde{A}_i$: $P(\tilde{A}_i < \tau_r)$ for $r \in \{0, 1, 2, \dots, D\}$. Substituting (6.a) for the latent variable makes it possible to explain the probabilities as a function of the distribution of the disturbance values: $P(\tilde{A}_i < \tau_r) = P((S_i + \tilde{\varepsilon}_i) < \tau_r) = P(\tilde{\varepsilon}_i < (\tau_r - S_i))$.

Standardizing the random term - by re-expressing its values in units of dispersion - and substituting (6.b) for S_i , we obtain:

$$P(\tilde{A}_i < \tau_r) = P\left(\frac{\tilde{\varepsilon}_i}{\sigma} < \left(\frac{\tau_r - S_i}{\sigma}\right)\right) = F\left(\rho_r - \sum_{m=1}^{m=M} \beta_m \cdot S_{i,m}\right),$$

where F is the standardized disturbance distribution function whose argument is specified by parameters defined only up to unidentifiable constants (ω_0 and σ). Therefore, letting:

$\rho_r = (\tau_r - \omega_0)/\sigma$, $\beta_m = \omega_m/\sigma$, and

$$S_i^* = \sum_{m=1}^{m=M} \beta_m \cdot S_{i,m}, \quad (8.a)$$

the response probabilities (7.a) to (7.c) are finally determined by:

$$\pi_{i,r} = F(\rho_{r+1} - S_i^*) - F(\rho_r - S_i^*), \quad \text{for } r \in \{1, 2, \dots, D-1\}, \quad (8.b)$$

$$\pi_{i,0} = F(\rho_1 - S_i^*) \quad (8.c) \quad \text{and} \quad \pi_{i,D} = 1 - F(\rho_D - S_i^*). \quad (8.d)$$

Figure 4 summarizes the model construction detailed above. Its properties are presented in Appendix I.

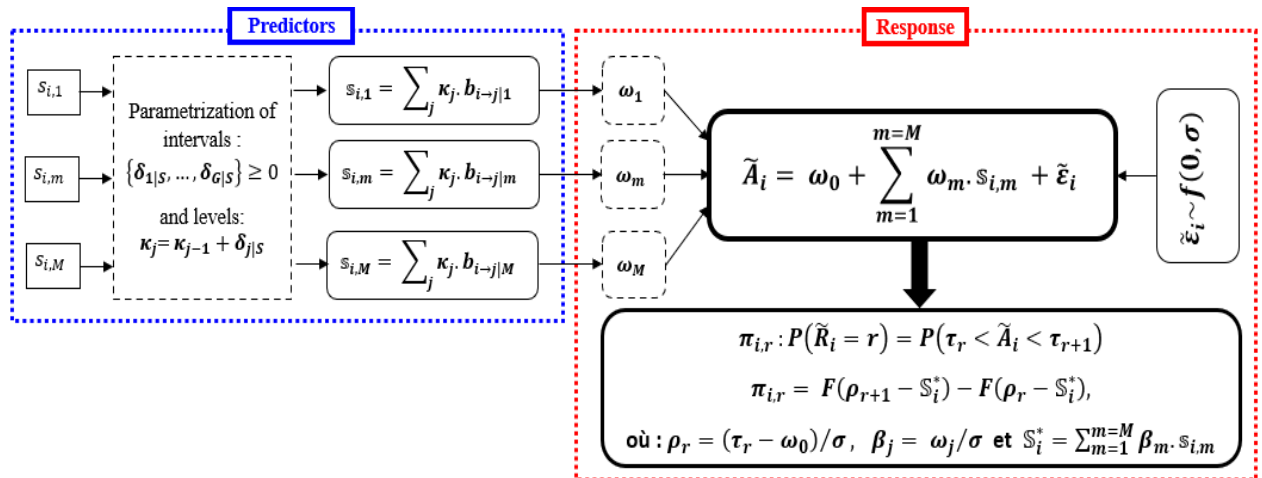


Figure 4. Mathematical formalization

Therefore, substituting (6.c) into (8.b, c, d) finally gives:

$$\pi_{i,r} = P(\tilde{R}_i = r) = P(\tilde{A}_i < \tau_{r+1}) - P(\tilde{A}_i < \tau_r) = [1 + e^{-(\rho_{r+1} - \mathbb{S}_i^*)}]^{-1} - [1 + e^{-(\rho_r - \mathbb{S}_i^*)}]^{-1}. \quad (9)$$

Figure 5 illustrates equation (9). Each of its quadrants corresponds to a pair of values assigned to the limits of the response category under consideration: $[\tau_r, \tau_{r+1}]$, or standardized: $[\rho_r, \rho_{r+1}]$.

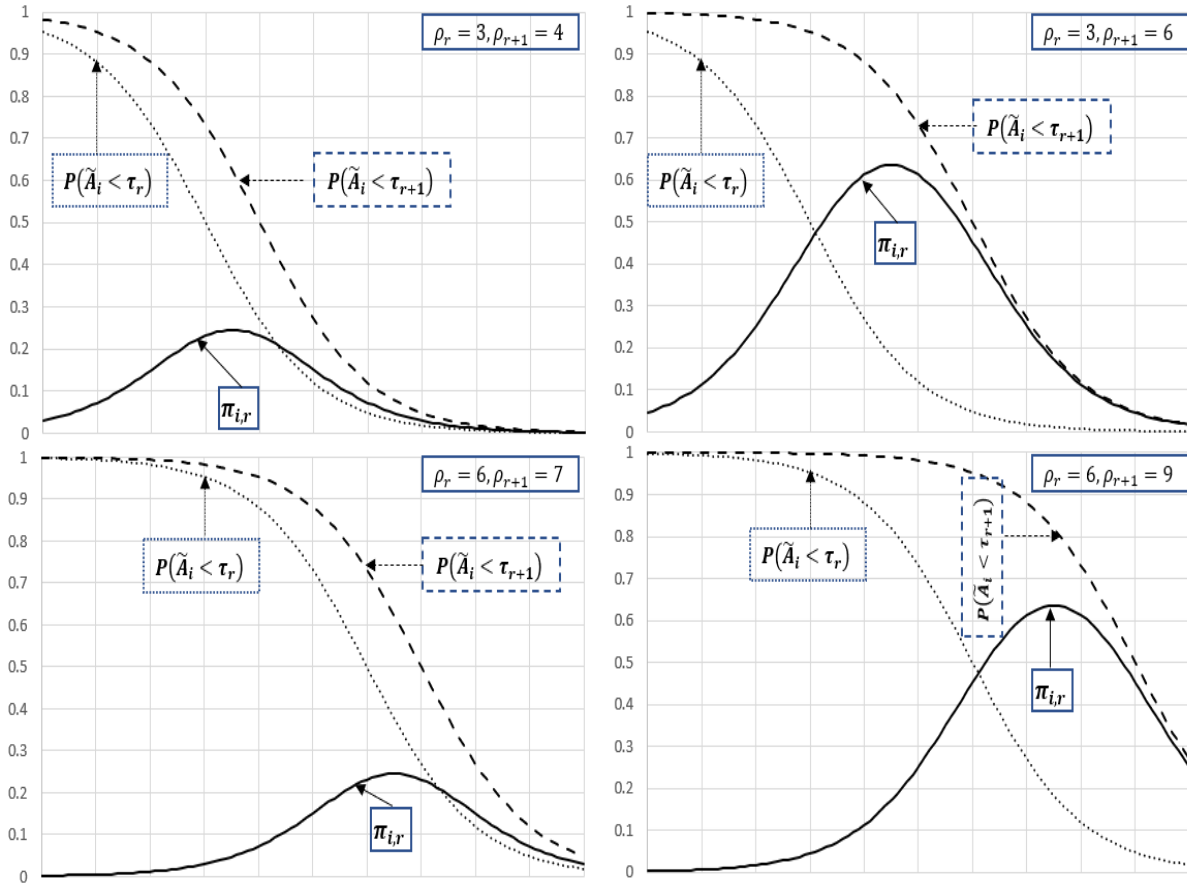


Figure 5. Decomposing the effects of $\mathbb{S}_i^*$ (horizontal axes) on the probabilities (vertical axes)

Each quadrant shows the effect of satisfaction, $\mathbb{S}_i^*$, plotted on the x-axis. The two curves taking an inverted S shape - represented by dotted and dashed lines - are decreasing because as satisfaction increases, the attitude becomes more favorable, and consequently the probabilities that it be above (below) the category bounds (thresholds) increase (decrease):

$$\mathbb{S}_i^* \nearrow \Rightarrow \tilde{A}_i \nearrow \Rightarrow \begin{cases} P(\tilde{A}_i > \tau_r) \nearrow \\ P(\tilde{A}_i > \tau_{r+1}) \nearrow \end{cases} \Rightarrow \begin{cases} P(\tilde{A}_i < \tau_r) \searrow \\ P(\tilde{A}_i < \tau_{r+1}) \searrow \end{cases}$$

The gap between the two limits, $\Delta_{r|R} = \rho_{r+1} - \rho_r$, determines the distance separating these two curves (formal connection explained in point I.3 of Annex I), which move to the right as these limits increase. The probability, $\pi_{i,r}$, that the consumer will express a response equal to r , is shown as a solid line. It results from the difference between the two inverted S curves

and thus takes the form of a symmetrical bell curve. The greater the difference between the thresholds, the sharper the maximum. The higher the thresholds, the further to the right this maximum will appear.

The model outlined is analogous to Torgerson's *law of categorical judgments* (1958, pp. 205-246), except that this law can predict negative probabilities (Rosner and Kochanski, 2009). It can only be better justified by the *latent trait theory* developed for scores from Likert scales, leading to the *graded-response logistic cumulative probability model* (Samejima, 1969). Unfortunately, it has not been widely applied in marketing. Singh et al (1990) make an exception: they exploit it to design an adaptive survey method that aims to expose respondents to only those items that are most revealing of their attitudes. Kau and Hill (1972) and Bettman (1974) also motivate the use of the binomial (probit) regression model by the hypotheses of a latent continuum and a critical threshold to explain a purchase decision, or the satisfaction, aroused by an attribute.

Estimation

The likelihood of the *responses* expressed by the sample of consumers is defined by the product of the probabilities of all possible *response* categories, each raised to a binary power indicative of the *response* actually expressed by each:

$$L = \prod_{i=1}^{i=N} \prod_{r=0}^{r=D} \pi_{i,r}^{B_{i \rightarrow r}}.$$

Thus, in a manner analogous to the encoding of predictor data, the exponents of the probabilities are defined by:

$$B_{i \rightarrow r} = \begin{cases} 1, & \text{si } \tilde{R}_i = r, \\ 0, & \text{si } \tilde{R}_i \neq r. \end{cases}$$

Optimal parameter estimates are obtained by maximizing L or its logarithm: $\mathcal{L} = \ln L$. Thus, they solve:

$$\text{MAX}_{\hat{\Theta}} \mathcal{L} = \left[\sum_{i=1}^{i=N} \sum_{r=0}^{r=10} B_{i \rightarrow r} \cdot \ln \hat{\pi}_{i,r} \right], \quad (10)$$

where $\hat{\Theta}$ represents the vector of parameter estimates, recapped in Table 3. Appendix II explains how we programmed (10) and calculated the precision of our *bootstrapping* estimates.

Table 3. Summary of the unknowns

Parameters: $\theta \in \Theta$	Count
Weighting the satisfaction criteria: $\beta_1, \dots, \beta_m, \dots, \beta_M$	M
Quantifying the $s_{i,m}$ en $s_{i,m} : \delta_{1 S}, \dots, \delta_{j S}, \dots, \delta_{G S}$	G
Determining the response thresholds: $\rho_1, \Delta_{1 R}, \dots, \Delta_{r R}, \dots, \Delta_{D-1 R}$	D

In addition to the coefficients reflecting the importance of the satisfaction criteria (β_j), Θ includes the intervals ($\delta_{j|S}$) between levels of the common satisfaction scale, as well as those between thresholds determining the degrees of response probabilities. The latter, too, must respect the monotonicity condition: $\rho_{r+1} \geq \rho_r$, resulting from the standardization of ordered thresholds ($\tau_{r+1} \geq \tau_r$). If this constraint were to be violated, then estimated values of $\pi_{i,r}$ would turn out to be negative¹⁸ and the calculation of $\mathcal{L}$ would be rendered impossible. This is why we proceeded, as for the quantification of predictors, by introducing positive increments ($\Delta_{r|R}$), specifying that¹⁹: $\rho_{r+1} = \rho_r + \Delta_{r|R}, \forall r \in \{1, 2, \dots, D-1\}$, with: $\Delta_{r|R} = \nabla_{r|R}^2$.

Implementation

Below, we demonstrate the practicability of our methodology with the collaboration of a firm competing in a European country *ICT* market. This company has entrusted us with data from its biannual satisfaction barometer, collected between 2019 and 2021 by market-research contractor: *IPSOS*. After an exploratory study, we selected two samples, each representative of the core customer base of two main competitors.: **C1** and **C2**. More specifically, we focused on the segment formed by patrons of each of these two companies for the three product-service categories each of them offers. Those clients are best able to make reliable overall judgments about their unique supplier and its services. The samples for **C1** and **C2** count **4,036** and **2,435** customers respectively. The information gathered covers

- their willingness to recommend their supplier (R):
from $r = 0$ meaning *certainly not* to $r = 10$ *certainly*,
- the four indicators of their satisfaction with
 - the functional performance of the three product-services:
digital TV (subscript: $DT \rightarrow s_{i,DT}$), fixed internet (subscript: $IT \rightarrow s_{i,IT}$)
and mobile telecoms (subscript: $MB \rightarrow s_{i,MB}$);
 - the quality-to-price ratio of all services provided (subscript: $QP \rightarrow s_{i,QP}$).

¹⁸ According to (7.a) and since every cumulative function F is increasing, $\hat{\rho}_{r+1} < \hat{\rho}_r \Rightarrow \hat{\pi}_{i,r} < 0$.

¹⁹ ρ_1 can take any value. Note that $\hat{\Delta}_{r|R}$ cannot be zero, because if it were, then: $\hat{\pi}_{i,r}$ would be zero, its $\log -\infty$ and the value to be maximized (10) could not be calculated.

The semantic differential scale for the measurement of satisfaction - common to all four criteria - is the 5-point grading defined in Table 2.

On each of the two samples, we tested the model described in Figure 4 linking the satisfaction scores: $\{S_{i,IT}, S_{i,MB}, S_{i,TV}, S_{i,QP}\}$, optimally quantified: $\{S_{i,IT}, S_{i,MB}, S_{i,TV}, S_{i,QP}\}$, to the probabilities that willingness to recommend of various degrees will be manifested: $\pi_{i,r} = P(\tilde{R}_i = r)$. Adding control covariates - such as customers' ownership of *ICT* devices, their involvement in decisions to purchase such equipment, their age, etc. - does not improve the goodness of fit and hardly alters the parameter estimates. Thus, using the inference method whose computer programming is described in Appendix II, we twice calibrated our cumulative logistic multinomial regression model for ordinal categorical predictors: $i \in \{1, \dots, 4.036\}$ for **C1** and $i \in \{1, \dots, 2.435\}$ for **C2**. Estimation - by random resampling and likelihood maximization of each of the 10,000 bootstrapped samples - yields the results shown in Figures 6 through 12.b and Tables 4 through 8, commented below. Here we distinguish each parameter θ (reported in Table 3) from the mean $\hat{\theta}$ of its estimated values:

$$\hat{\theta} = \sum_{b=1}^{b=10.000} \hat{\theta}^{(b)} / 10.000,$$

$\hat{\theta}^{(b)}$ denoting the estimate of θ derived from the b^{th} bootstrapped sample.

Relative importance of the attributes of the offering

The estimated effects of the consumer ratings of the service line components are all positive, as shown by their distributions in Figure 6. None of the eight histograms include zero. The $\hat{\beta}_m^{(b)}$ are all positive, so the effects of the four satisfaction motives on the probabilities of recommending operators are undeniable. Table 4 confirms that all estimators are highly accurate. The functional performance of mobile telephony proves to be the most decisive for both **C2** and **C1**: the distribution of $\hat{\beta}_{MB}^{(b)}$ stands out to the right of the other three, without overlapping with those of $\hat{\beta}_{DT}^{(b)}$ and $\hat{\beta}_{QP}^{(b)}$. Although the minima of $\hat{\beta}_{MB}^{(b)}$ are lower than the maxima of $\hat{\beta}_{IT}^{(b)}$, the distributions of their differences, $\hat{\beta}_{MB}^{(b)} - \hat{\beta}_{IT}^{(b)}$, shown in Figure 7, clearly prove that²⁰: $\beta_{MB} > \beta_{IT}$.

²⁰ Just comparing the positions of the estimators' distributions is not enough, since it does not take into account their covariance, which we do not have to worry about when we rely directly on the distribution of their difference.

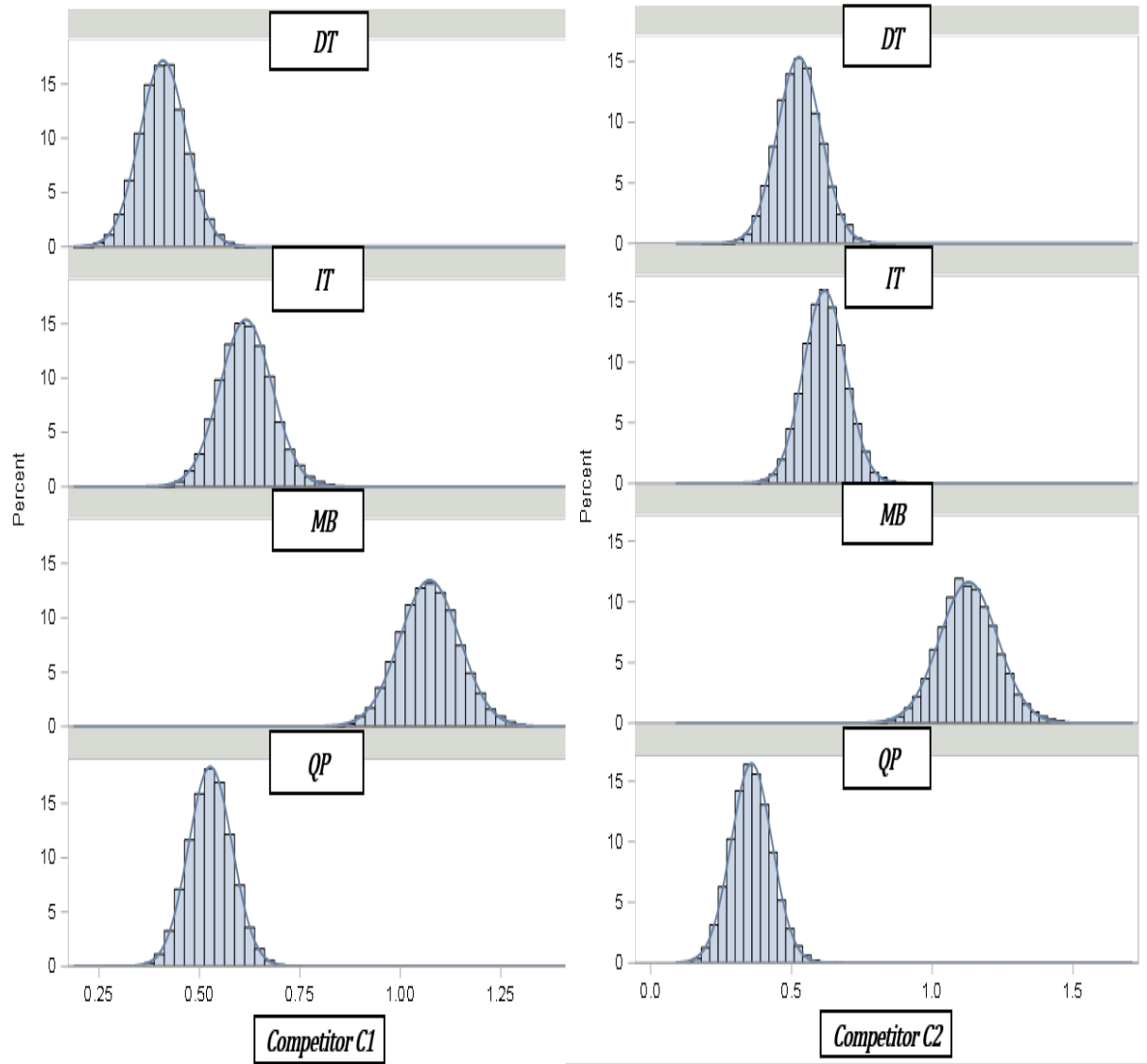


Figure 6. Distributions of satisfaction criteria coefficient estimates

Table 4. Characteristics of distributions of satisfaction criteria estimated weights

Statistics	Competitor C1				Competitor C2			
	$\hat{\beta}_{DT}$	$\hat{\beta}_{IT}$	$\hat{\beta}_{MB}$	$\hat{\beta}_{QP}$	$\hat{\beta}_{DT}$	$\hat{\beta}_{IT}$	$\hat{\beta}_{MB}$	$\hat{\beta}_{QP}$
Mean (<i>M</i>)	0.409	0.616	1.073	0.527	0.527	0.618	1.129	0.358
Standard deviation (<i>SD</i>)	0.058	0.065	0.074	0.054	0.078	0.075	0.103	0.073
Precision (<i>M/SD</i>)	7.036	9.482	14.481	9.704	6.768	8.206	10.977	4.926
Minimum	0.188	0.387	0.821	0.285	0.193	0.345	0.794	0.111
Maximum	0.617	0.900	1.405	0.752	0.866	0.937	1.698	0.679

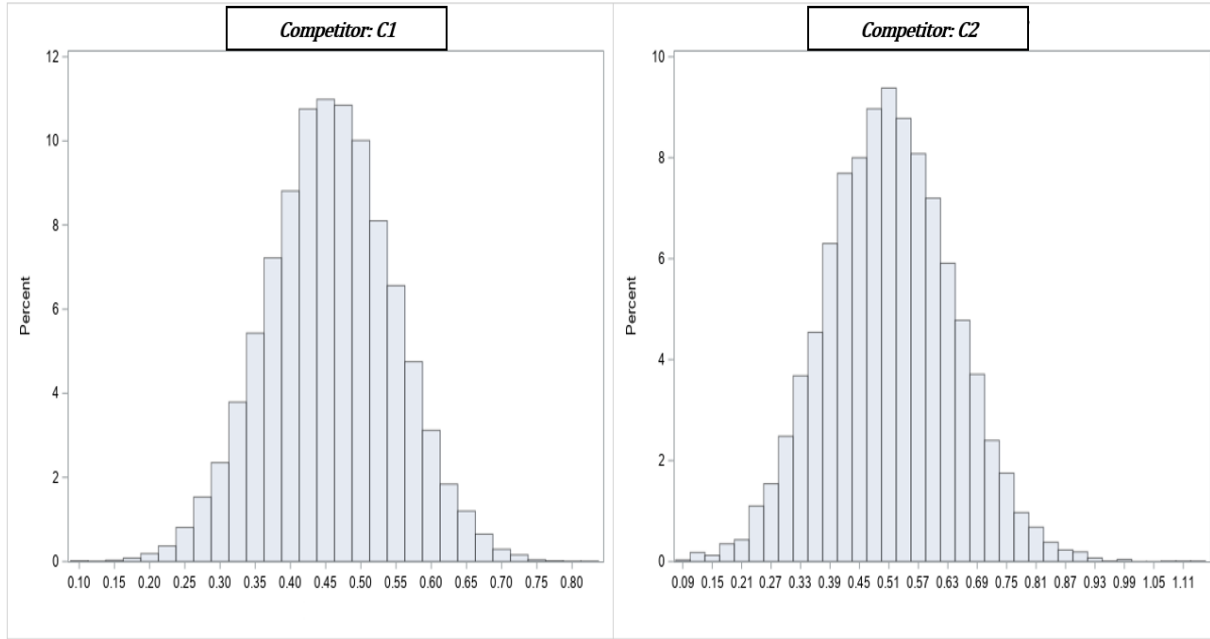


Figure 7. Test of comparison between two coefficients: distributions of $(\hat{\beta}_{MB}^{(b)} - \hat{\beta}_{IT}^{(b)})$

Overall Predictive Power

Among the various rank correlation measures suitable for studying associations between ordinal variables, we chose Goodman-Kruskal's Γ (gamma), which can be interpreted as an ordinary correlation coefficient: it ranges from -1 to $+1$ (see: **Goodness of fit** section in Appendix III). Figure 8 confirms the goodness of fit of the model to the data, since the values of Γ are all clearly positive, and for both competitors: the minima are indeed 0.551 for **C1** and 0.571 for **C2**.

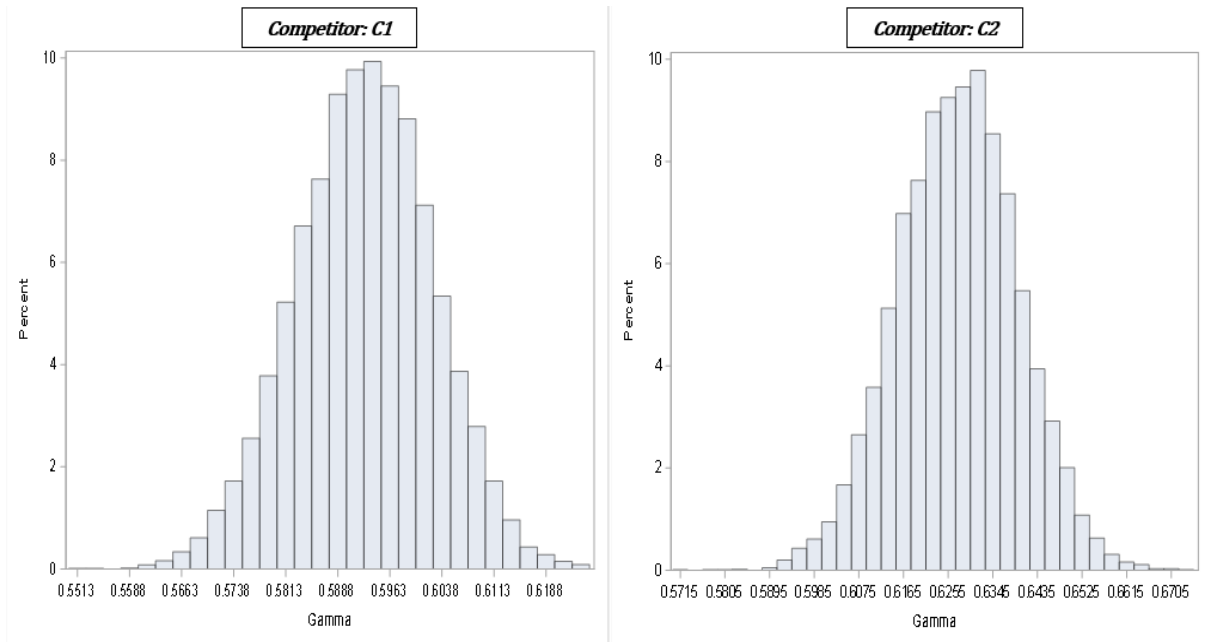


Figure 8. Distributions of estimated values of the rank correlation coefficient Γ

Optimal calibration of satisfaction levels

In the following figures, the parallelization of box plots facilitates contrasts between the distributions of the estimators to be compared (for instructions on how to interpret them, please refer to the section Graphical Representations in Appendix III). Figure 9 depicts vertically the distributions of the quantified score estimates, $\kappa_{j|S}$, of the semantic satisfaction scale, with fixed anchor points: $\kappa_{0|S} = 0$ and $\kappa_{4|S} = 4$. The two curves show the same pattern, that of a very stretched S-shape for **C1**, but with a slight rebound to the top rung for **C2**. To be more precise,

- (a) The five categories appear to be quite distinct from each other: Table 5 shows that all $\hat{\delta}_{j|S}$ are sufficiently larger than **0** to assume that consumers differentiate the five consecutive satisfaction levels, since the smallest standardized difference (**C2**: $\hat{\delta}_{1|S}$) exceeds 3.
- (b) Most of the distances between levels are of the order of unity. Some are smaller: for **C1**, $\hat{\delta}_{3|S}$, and $\hat{\delta}_{4|S}$, and for **C2**, $\hat{\delta}_{1|S}$ and $\hat{\delta}_{3|S}$; for **C1** they are more than two standard deviations smaller than 1, which is not the case for **C2**.
- (c) The estimator for the first interval, $\delta_{1|S}$: between **0** for *not satisfied at all* and **1** for *not quite satisfied*, is the least precise, while the values of the third and fourth are better discerned.
- (d) A differential more clearly distinguishes *satisfied* ($j = 2$) from *not quite satisfied* ($j = 1$): $\hat{\delta}_{2|S}$ is larger than the others. However, the hypothesis that $\delta_{2|S}$ is equal to 1 cannot be rejected: the means of $\hat{\delta}_{2|S}^{(b)}$ are only less than two standard deviations above **1**: 1.77 and 1.27, respectively, for **C1** and **C2**. A priori, we would have expected a much steeper jump, given the semantic leap.

Table 5. Differentiation of satisfaction levels: $\hat{\delta}_{j|S}^{(b)} = \hat{\kappa}_j^{(b)} - \hat{\kappa}_{j-1}^{(b)}$
Standardized difference = ratio of the mean of the differences to their standard deviation

COMPETITORS	STATISTICS	$\hat{\delta}_{1 S}$	$\hat{\delta}_{2 S}$	$\hat{\delta}_{3 S}$	$\hat{\delta}_{4 S}$
C1	Mean of the differences	1.083	1.213	0.847	0.856
	Standard deviation of the differences	0.195	0.120	0.062	0.064
	Standardized difference	5.555	10.097	13.698	13.477
C2	Mean of the differences	0.854	1.188	0.906	1.053
	Standard deviation of the differences	0.256	0.148	0.081	0.090
	Standardized difference	3.335	8.003	11.205	11.637

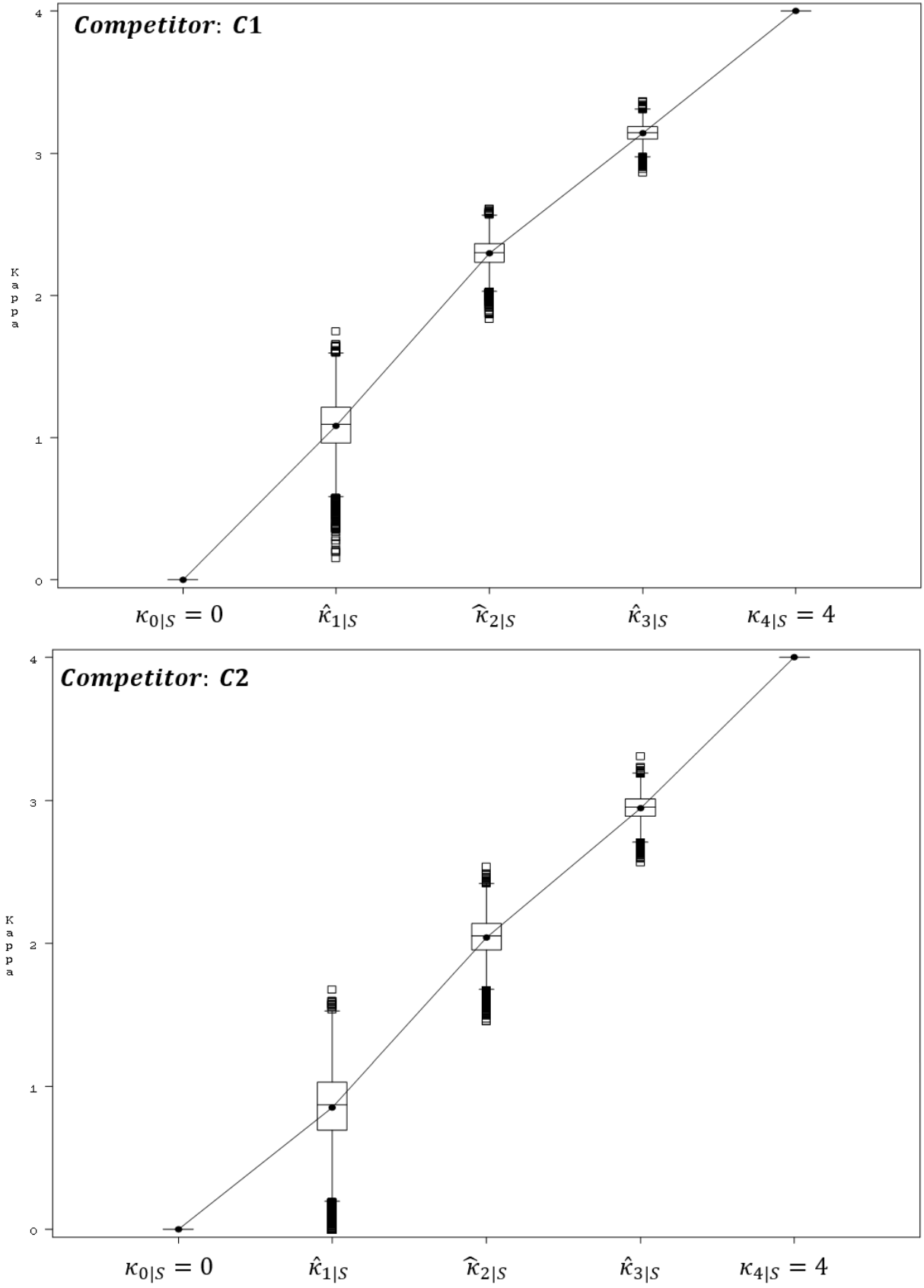


Figure 9. Distributions of estimates to be assigned to the degrees, j , of satisfaction: $\hat{\kappa}_{j|S}^{(b)}$

Our simulations suggest that the satisfaction scale is very close to linear. To support this conclusion, we tested the cumulative logistic regression model on quantitative predictors using the same 10,000 bootstrapped samples. In doing so, we equalized all intervals between steps of the satisfaction scale. We therefore set: $\delta_{j|S} = 1, \forall j$. From these additional analyses, we derived the distributions of Γ values shown in Figure 10. These and those in Figure 8 are virtually superimposable. We can therefore conclude that treating the ordinal categorical satisfaction scores as quantitative in no way worsens the model's fit to the data. Table 6 completes the sensitivity analysis by examining the effect on the estimated regression coefficients.

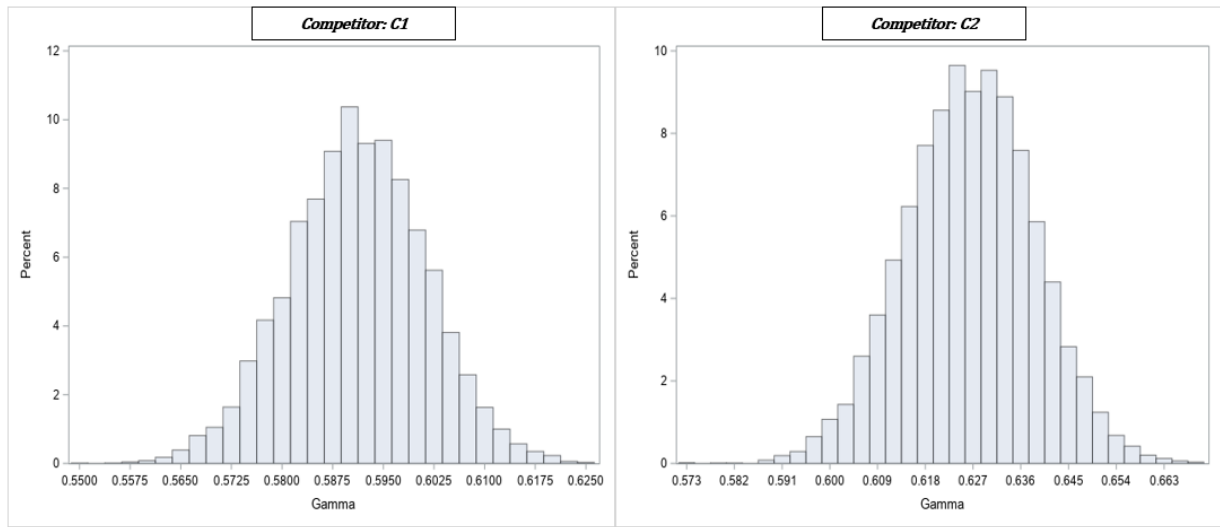


Figure 10. Histograms of estimated Γ values reflecting the quality of the simplified model

Table 6. Sensitivity of service-components weights

Standard deviations are shown in brackets. For ease of comparison, the **ordinal** section is copied from Table 4.

Satisfaction scale	Competitor C1				Competitor C2			
	$\hat{\beta}_{DT}$	$\hat{\beta}_{IT}$	$\hat{\beta}_{MB}$	$\hat{\beta}_{QP}$	$\hat{\beta}_{DT}$	$\hat{\beta}_{IT}$	$\hat{\beta}_{MB}$	$\hat{\beta}_{QP}$
Ordinal	0.409 (0.058)	0.616 (0.065)	1.073 (0.074)	0.527 (0.054)	0.527 (0.078)	0.618 (0.075)	1.129 (0.103)	0.358 (0.073)
Interval ($\delta_{j S} = 1, \forall j$)	0.384 (0.050)	0.572 (0.047)	0.961 (0.055)	0.506 (0.043)	0.512 (0.068)	0.613 (0.066)	1.115 (0.086)	0.362 (0.060)

Assimilating the satisfaction scale to an interval scale does not alter the prioritization of the components of the product-service offer: the estimates of their coefficients change little; their standard deviations are reduced by the simplification of the model (the $\delta_{j|S}$ are predetermined). However, we must be careful not to conclude that the same will apply in other cases. So much so that based on Table 5 and Figure 9, we could have considered that some intervals are less than 1, i.e. for **C1** : $\delta_{3|S}$ and $\delta_{4|S}$, and for **C2** : $\delta_{1|S}$ and $\delta_{3|S}$.

Response thresholds

The ρ_r parameters, apart from a linear transformation (standardization: division by σ), are nothing more than the limits of the intervals that cut the range of variation of the latent variable underlying the willingness to recommend. The differences between them allow us to test whether the amplitudes of the deviations between these response thresholds are

- comparable, in which case the propensity to recommend could be regarded as a quantitative scale (interval or even ratio),
- negligible, in which case certain response categories could be merged, justifying performance diagnostics based on simple *NPS*-type scores.

Relevant statistics. The distributions of the estimates of category boundaries are summarized in Figure 11, which reveals the general shape of the curves connecting the means of the $\hat{\rho}_r$ -distributions. Far from being linear, it appears to be convex, *J*-shaped. Complementarily, Table 7 highlights that the estimated length of the intervals, $\hat{\Delta}_{r|R}$, tends to grow markedly from $\hat{\rho}_4$ to $\hat{\rho}_5$, and then beyond $\hat{\rho}_6$ (for **C1**, this growth slows slightly down between $\hat{\rho}_9$ and $\hat{\rho}_{10}$).

Long, unequal and widening intervals. The last column of Table 7 shows how statistically different the recommendation thresholds are. In units of standard deviations, the $\hat{\Delta}_{r|R}$ are not only markedly greater than zero, but also unequal. Thus, the Reichheld scale cannot be equated with an interval scale, and the *NPS* is untenable. There is no doubt that the intervals are such that any grouping of response categories is prohibited, even more so when the willingness to recommend is strong ($R \geq 6$). Such mergers are prohibited because the higher the response thresholds, the lower the probability of overstepping them: the risk increases that the attachment to the brand is insufficient to induce a greater willingness to recommend it.

Table 8 shows the values of the interval comparison statistics estimated from the two samples:

$$\mathbb{d}_{r|R} = \left(\hat{\Delta}_{r|R}^{(C1)} - \hat{\Delta}_{r|R}^{(C2)} \right) / \sqrt{\left(\hat{\sigma} \left[\hat{\Delta}_{r|R}^{(C1)} \right] \right)^2 + \left(\hat{\sigma} \left[\hat{\Delta}_{r|R}^{(C2)} \right] \right)^2},$$

where superscripts identify clienteles. The $\mathbb{d}_{r|R}$ confirm that the results for **C1** and **C2** are roughly similar, except for: $\hat{\Delta}_{2|R}$, $\hat{\Delta}_{7|R}$ et $\hat{\Delta}_{9|R}$.

Behind the numbers: intentions. To illustrate the importance of thresholds, we have graphically studied the impact of their spread along the response scale on the distribution of response probabilities: $\pi_{i,r}$. Their effect, demonstrated in Figure 5, is difficult to assess analytically (we do so in §I.3 of Appendix I). Therefore, we have evaluated these intention probabilities by applying formula (9) for each of the estimated intervals $[\hat{\rho}_r, \hat{\rho}_{r+1}]$, as a

function of overall satisfaction: $\mathbb{S}_i^*$. The results obtained in this way - for $\hat{\rho}_r$ taken from sample **C1** (means of the distributions in the upper part of Figure 11)²¹ - are shown in Figure 12.a. The modes of the functions $\hat{\pi}_{i,r}(\mathbb{S}_i^*)$ are ordered from left to right, and the growth of the maxima mirrors that of the J-curve commented on above. $\hat{\pi}_{i,3}$, $\hat{\pi}_{i,4}$ and $\hat{\pi}_{i,6}$ are exceptions because $\hat{\Delta}_{3|R}$, $\hat{\Delta}_{4|R}$ and $\hat{\Delta}_{6|R}$ flag trend breaks in Figure 11.

Table 7. Statistics for evaluating intervals between thresholds

Intervals: $\hat{\Delta}_{r R}$	C1		
	Estimate	Standard deviation: $\hat{\sigma}[\hat{\Delta}_{r R}]$	$z(\hat{\Delta}_{r R}) = \hat{\Delta}_{r R}/\hat{\sigma}[\hat{\Delta}_{r R}]$
$\hat{\Delta}_{1 R} = \hat{\rho}_2 - \hat{\rho}_1$	0.674	0.198	3.401
$\hat{\Delta}_{2 R} = \hat{\rho}_3 - \hat{\rho}_2$	1.080	0.177	6.086
$\hat{\Delta}_{3 R} = \hat{\rho}_4 - \hat{\rho}_3$	0.519	0.088	5.899
$\hat{\Delta}_{4 R} = \hat{\rho}_5 - \hat{\rho}_4$	0.617	0.080	7.703
$\hat{\Delta}_{5 R} = \hat{\rho}_6 - \hat{\rho}_5$	1.309	0.079	16.544
$\hat{\Delta}_{6 R} = \hat{\rho}_7 - \hat{\rho}_6$	0.943	0.050	18.787
$\hat{\Delta}_{7 R} = \hat{\rho}_8 - \hat{\rho}_7$	1.459	0.049	30.050
$\hat{\Delta}_{8 R} = \hat{\rho}_9 - \hat{\rho}_8$	1.965	0.050	39.248
$\hat{\Delta}_{9 R} = \hat{\rho}_{10} - \hat{\rho}_9$	1.893	0.060	31.506
Intervals: $\hat{\Delta}_{r R}$	C2		
	Estimate	Standard deviation: $\hat{\sigma}[\hat{\Delta}_{r R}]$	$z(\hat{\Delta}_{r R}) = \hat{\Delta}_{r R}/\hat{\sigma}[\hat{\Delta}_{r R}]$
$\hat{\Delta}_{1 R} = \hat{\rho}_2 - \hat{\rho}_1$	1.160	0.355	3.272
$\hat{\Delta}_{2 R} = \hat{\rho}_3 - \hat{\rho}_2$	0.435	0.152	2.861
$\hat{\Delta}_{3 R} = \hat{\rho}_4 - \hat{\rho}_3$	0.645	0.147	4.381
$\hat{\Delta}_{4 R} = \hat{\rho}_5 - \hat{\rho}_4$	0.454	0.095	4.762
$\hat{\Delta}_{5 R} = \hat{\rho}_6 - \hat{\rho}_5$	1.276	0.115	11.069
$\hat{\Delta}_{6 R} = \hat{\rho}_7 - \hat{\rho}_6$	1.009	0.073	13.744
$\hat{\Delta}_{7 R} = \hat{\rho}_8 - \hat{\rho}_7$	1.602	0.067	23.994
$\hat{\Delta}_{8 R} = \hat{\rho}_9 - \hat{\rho}_8$	2.056	0.068	30.349
$\hat{\Delta}_{9 R} = \hat{\rho}_{10} - \hat{\rho}_9$	2.145	0.086	24.986

Table 8. Standardized differences between estimated intervals: **C1** versus **C2** clienteles

$\mathbb{d}_{1 R}$	$\mathbb{d}_{2 R}$	$\mathbb{d}_{3 R}$	$\mathbb{d}_{4 R}$	$\mathbb{d}_{5 R}$	$\mathbb{d}_{6 R}$	$\mathbb{d}_{7 R}$	$\mathbb{d}_{8 R}$	$\mathbb{d}_{9 R}$
-1.197	2.760	-0.738	1.315	0.241	-0.736	-1.735	-1.090	-2.402

²¹ For **C2**, the interpretation of the corresponding graph, which we do not display, leads to similar conclusions.

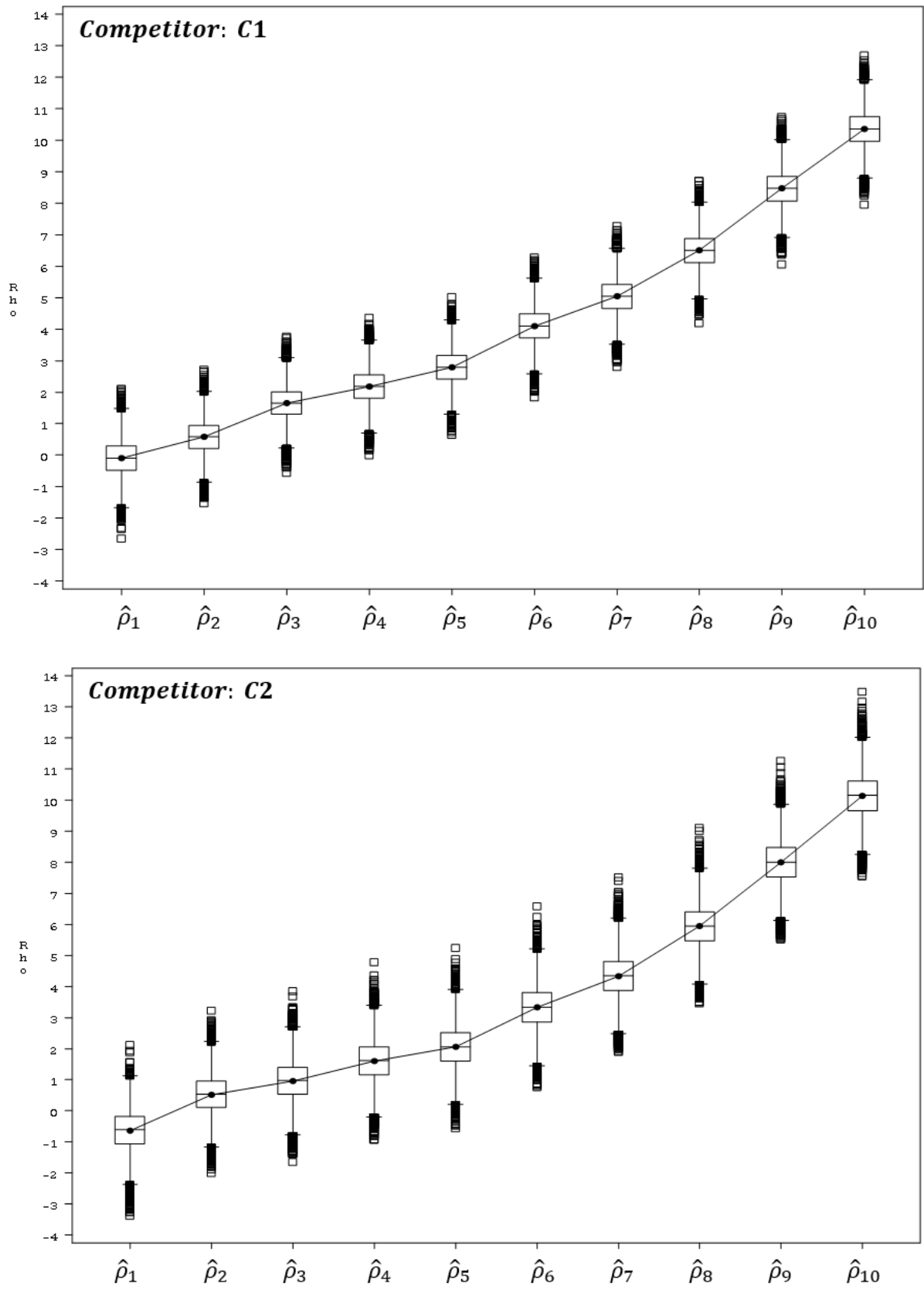


Figure 11. Distributions of estimated response thresholds: $\hat{\rho}_r^{(b)}$

To assess whether the R -scores form an interval scale, we smoothed the $\hat{\Delta}_{r|R}$ to make these gaps identical to one other²² and re-estimated the $\pi_{i,r}$. Despite the quality of this smoothing, its upshot is spectacular: the recalculated functions are shown in Figure 12.b. Each of them (except for the extremes) results from an invariant translation of the one to its left: the modes are uniformly scattered and the corresponding maximum probabilities are identical. The contrast between Figures 12.a and 12.b highlights the systematic errors in predicting overall satisfaction effects of satisfaction that would be induced by treating the measure of willingness to recommend as quantitative: here, underestimation of $\pi_{i,5}, \pi_{i,7}, \pi_{i,8}$ and $\pi_{i,9}$, and overestimation of $\pi_{i,3}, \pi_{i,4}, \pi_{i,6}$ and $\pi_{i,10}$. Obviously, the degrees of willingness to recommend cannot be interpreted as consecutive positive integers.

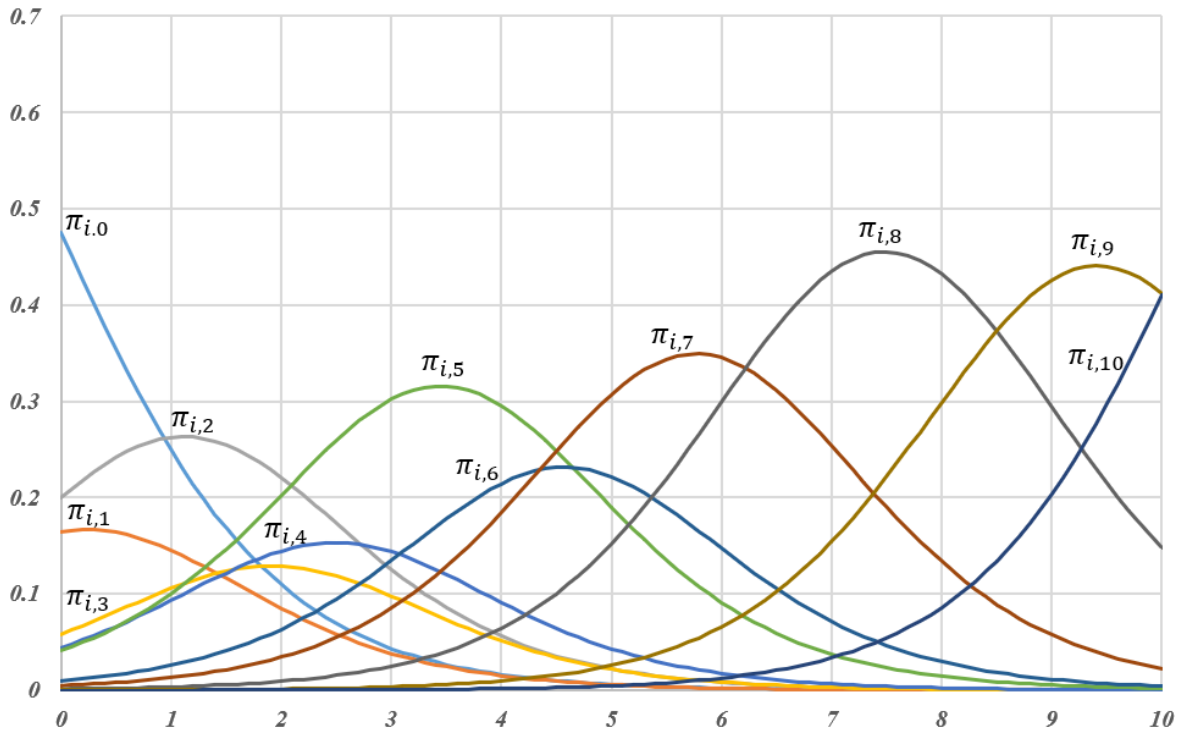


Figure 12.a. Effect of satisfaction (S_i^* along x-axis) on response probabilities (y-axis)²³, estimated using thresholds derived from **C1**.

²² More precisely, we have substituted the optimal estimates of ρ_r for their approximations by linear regression (coefficient of determination: 0.96).

²³ From the distributions of $\hat{\rho}_r$, we can straightforwardly derive confidence contours (percentiles) around the $\hat{\pi}_{i,r}(S_i^*)$ curves.

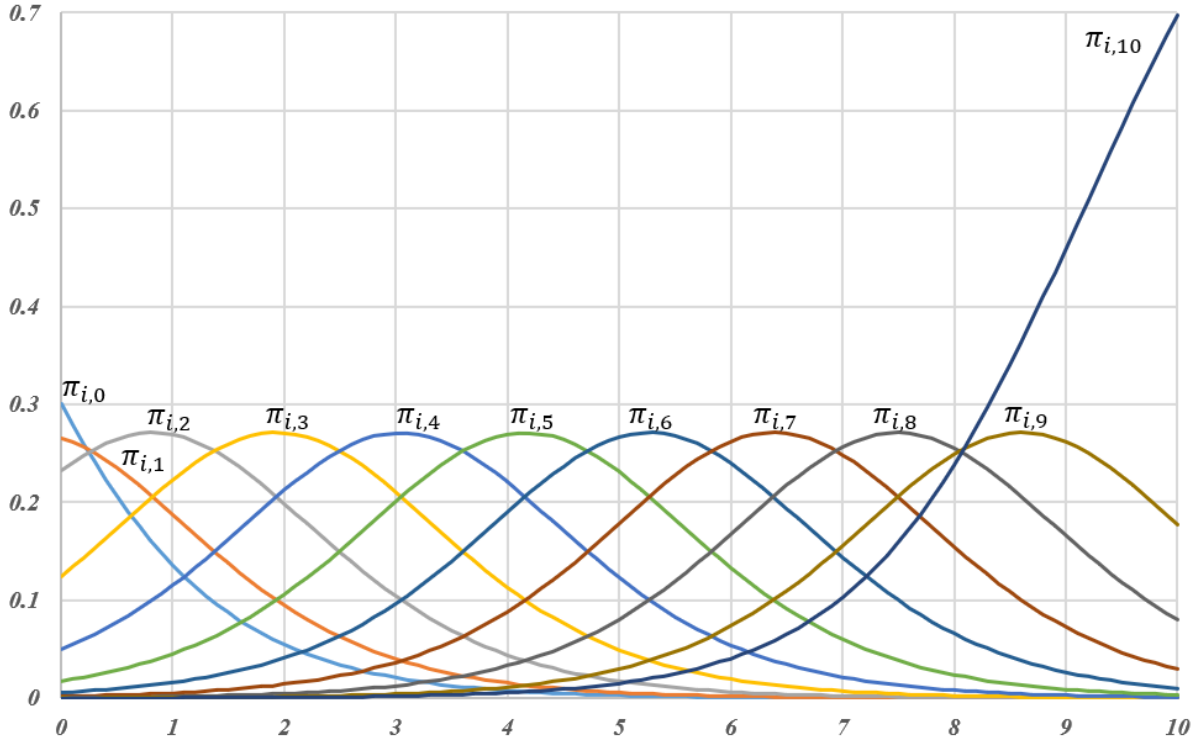


Figure 12.b. Response probabilities re-estimated using equidistant thresholds best correlated with the optimal $\hat{\rho}_r$ on which Figure 12.a has been built

Nonlinearity: a necessity. Little (1970) taught us that for a model to be used, it must be as easy to understand as possible... But only if it's "*robust*", meaning that when pushed to its extremes (minimum/maximum satisfaction scores) it still yields reasonable predictions. This is why we rejected a priori the specification (5) that linearly relates the quantified satisfaction scores ($s_{i,m}$) to the quantified response ($\mathbb{R}_i$). Indeed, there is a high risk of extrapolating, from (5), estimates that exceed the limits of the response range, or that these limits may never be reached. We have therefore specified the model defined by equation (9) to produce consistent predictions of response probabilities. However, while rigor is invaluable, it comes at a cost: that of complexity. This prompted us to present the effect of overall satisfaction in Figures 5, 12.a and 12.b, which can be easily integrated into an interactive decision support interface. However, making the nonlinear model more comprehensible does not relieve us from probing whether the linear model (5) might not be a credible approximation of reality.

To get to the heart of the matter, we calibrated it by maximizing the coefficient of determination ($\mathcal{R}^2$) of the quantified dependent variable ($\mathbb{R}_i$) by the linear combination of the effects of the multiple, also quantified, evaluations of the offer components ($\widehat{\mathbb{R}}_i$). For each of the two client groups, we therefore solved the following problem on 10,000 additional bootstrapped samples (using a routine analogous to that described in Table 12):

$$\underset{\{\hat{\varphi}_0, \hat{\varphi}_{DT}, \hat{\varphi}_{MB}, \hat{\varphi}_{IT}; \hat{\delta}_{1|R}, \dots, \hat{\delta}_{10|R}; \hat{\delta}_{1|S}, \dots, \hat{\delta}_{4|S}\}}{\text{MAX}} \mathcal{R}^2(\mathbb{R}_i, \widehat{\mathbb{R}}_i) \quad (11. a)$$

where $\mathcal{R}$ is the multiple correlation between

- on the one hand, the quantified dependent variable:

$$\mathbb{R}_i = \sum_{r=0}^{r=10} \hat{\kappa}_{r|R} \cdot B_{i \rightarrow r}, \text{ where: } \hat{\kappa}_{r|R} = \hat{\kappa}_{r-1|R} + \hat{\delta}_{r|R}, \text{ and such that: } \hat{\kappa}_{0|R} = 0 \text{ and } \hat{\kappa}_{10|R} = 10, \quad (11. b)$$

- on the other hand, the mix of the estimated effects of the jointly quantified predictors:

$$\widehat{\mathbb{R}}_i = \hat{\varphi}_0 + \sum_{m \in \{DT, IT, MB, QP\}} \hat{\varphi}_m \cdot \hat{s}_{i,m}, \text{ where: } \hat{s}_{i,m} = \sum_{j=0}^{j=4} \hat{\kappa}_{j|S} \cdot b_{i \rightarrow j|m}, \quad (11. c)$$

with: $\hat{\kappa}_{j|S} = \hat{\kappa}_{j-1|S} + \hat{\delta}_{j|S}$, and such that: $\hat{\kappa}_{0|S} = 0$ and $\hat{\kappa}_{4|S} = 4$.

The two times ten thousand solutions obtained in this way are summarized in Table 9.

Table 9. Main results from fitting the constrained linear response model

$$\widehat{\mathbb{R}}_{\text{max}} = \hat{\varphi}_0 + \left(4 \times \sum_{m \in \{DT, IT, MB, QP\}} \hat{\varphi}_m \right)$$

Clientele	Statistics	$\hat{\varphi}_0$	$\hat{\varphi}_{DT}$	$\hat{\varphi}_{IT}$	$\hat{\varphi}_{MB}$	$\hat{\varphi}_{QP}$	$\mathcal{R}^2$	$\widehat{\mathbb{R}}_{\text{max}}$	% de $R_i = 10$
C1	M	0.257	0.319	0.522	0.920	0.417	0.451	8.968	9.514%
	SD	0.810	0.055	0.066	0.100	0.056	0.014		
	M/SD	0.318	5.853	7.884	9.193	7.478	32.106		
C2	M	0.463	0.487	0.485	0.931	0.261	0.489	9.122	10.349%
	SD	0.771	0.094	0.074	0.102	0.064	0.017		
	M/SD	0.601	5.195	6.577	9.159	4.087	28.711		

M: Mean. **SD**: Standard Deviation. **M/SD**: Precision.

The last two columns confirm the inadequacy of the linear model: while around 10% of respondents (last column) express a likelihood-to-recommend score of 10, the model cannot predict a score higher than 9 (penultimate column). Clearly, the linear model does not pass Little's (2004: 1858) test of validity: what company could endorse a model that predicts that none of its highly satisfied customers would declare themselves certain to recommend it? If, notwithstanding its Achilles' heel, we were to rely on the linear model (11. b)-(11. c) to assess the relevance of the *NPS*, this score would still not be justified, as can be seen from Figures 13 and 14 materializing the calibration: $R \Rightarrow \mathbb{R}$.

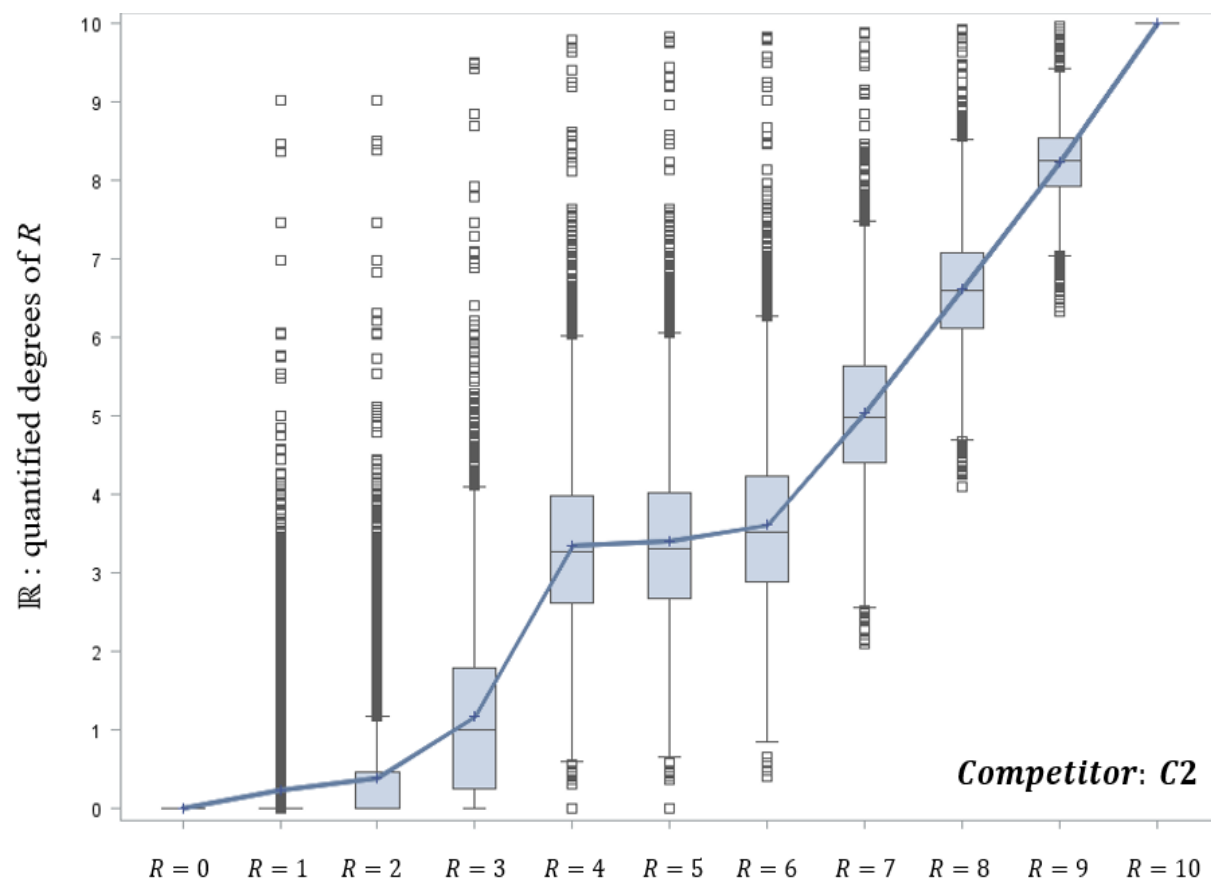
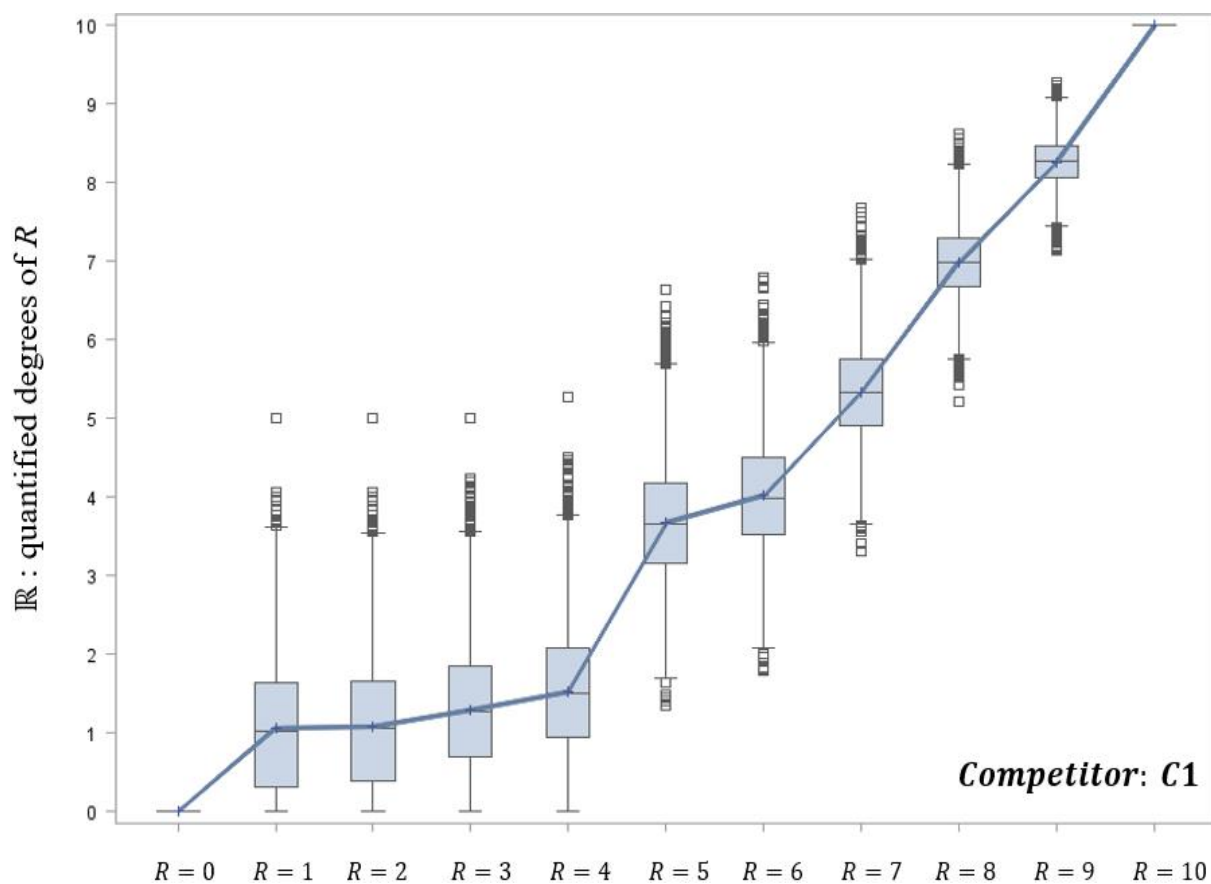


Figure 13. Distributions of quantified degrees of willingness-to-recommend

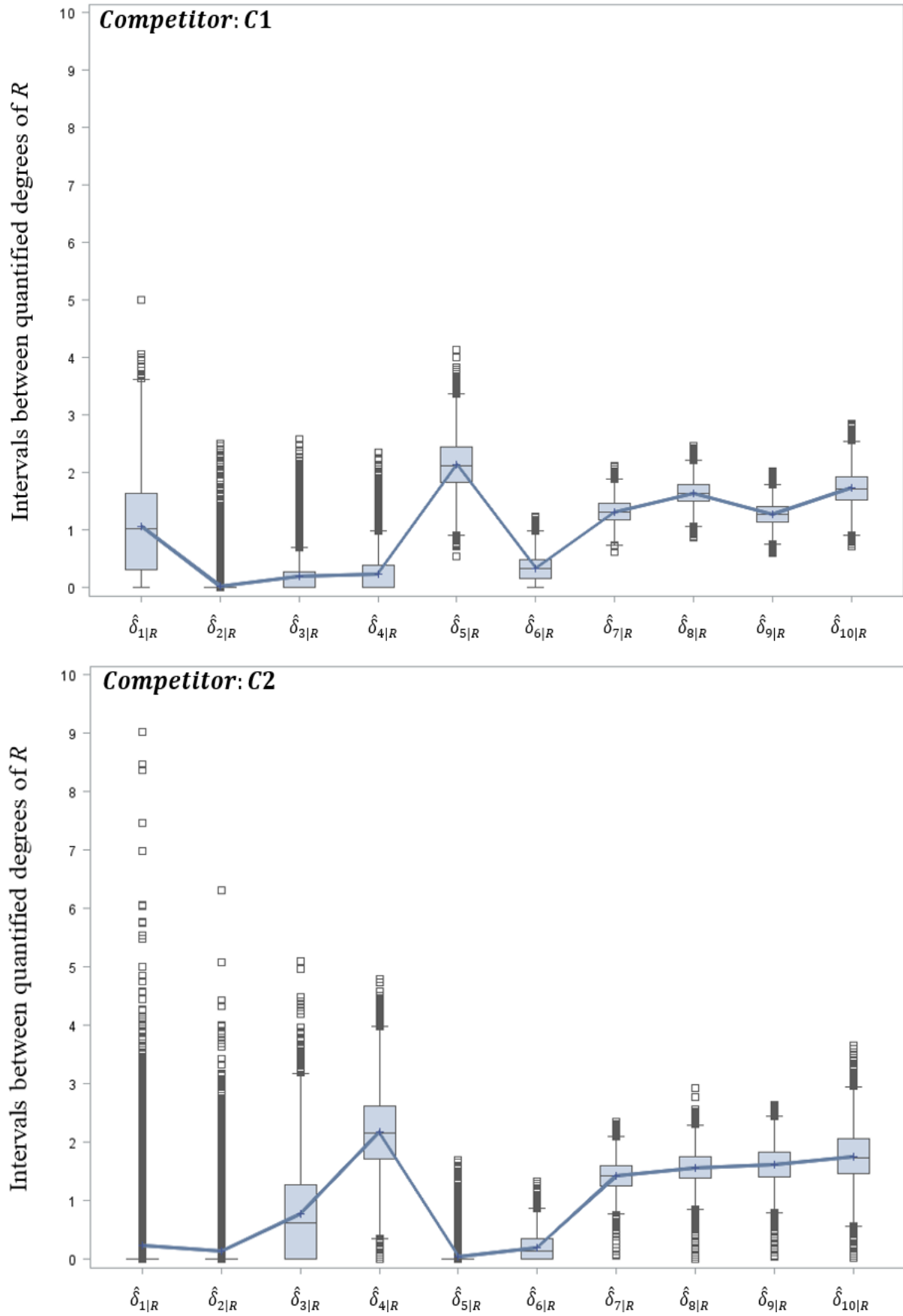


Figure 14. Distributions of differences between estimates of R -degrees

The curves in Figure 13 also exhibit a convex *J*-shape, but with two very marked plateaus of equivalent scores,

- for **C1**, in $R = \{1 \cup 2 \cup 3 \cup 4\}$, then in $\{5,6\}$ which is confirmed by Figure 14: the distributions of $\hat{\delta}_{2|R}$, $\hat{\delta}_{3|R}$, $\hat{\delta}_{4|R}$ and $\hat{\delta}_{6|R}$ are concentrated around zero;
- for **C2**, in $R = \{0 \cup 1 \cup 2\}$ and then in $\{4 \cup 5 \cup 6\}$, which is confirmed by Figure 14: the distributions of $\hat{\delta}_{1|R}$, $\hat{\delta}_{2|R}$, $\hat{\delta}_{5|R}$ and $\hat{\delta}_{6|R}$ also appear close to zero, but count a number of exceptional values.

On the other hand, the categories 7, 8, 9 and 10 are clearly distinct. Irrespective of the specification used, linear or not, our results therefore do not support the grouping of degrees 0 to 6, nor the clustering of the 9th and 10th, both of which Reichheld advocates since they condition the calculation of the *NPS*.

Specific features of the empiric findings

In the two cases studied, **C1** and **C2**, the 0-to-4 ratings on the satisfaction criteria can be interpreted as measurements on an interval scale²⁴. On the contrary, the optimal quantification of the response levels does not lead to equidistant levels. Furthermore, we have just shown that the mere extension of monotonic regression cannot guarantee credible diagnoses.

Our deductions are grounded not only on statistics, but also and above all on the graphical exploitation of the properties that bridge the gap between the thresholds that discretize latent attitudes - to which one pays too little attention²⁵ - and the response probabilities (see Figure 5). This visualization helped the firm we worked with to take a fresh look at the predictions of the effects of variations in satisfaction: specifically, at the redistribution of response probabilities in Figure 12.a. The company learned a great deal from where we ended up. They found the precision of the estimates of the coefficients of the satisfaction levers to be more than sufficient to revisit their prioritization. The relationships we have established between satisfaction and willingness to recommend are stronger than those reported in the literature: for example, the coefficient of determination obtained by Huang and Sudhir (2021, Table 4, col. IV: 324) is 0.20, against R^2 -values more than double for those obtained using the monotone regression model (11): see Table 9. This superiority might only be apparent, as the common method bias and halo effect might have inflated the value of the indicator of the

²⁴ However, we believe that this is due to the small number of satisfaction scale-points (5): had been there 11 (from 0 to 10), the optimally quantified satisfaction levels would probably not have aligned as shown in Figure 9.

²⁵ “Although we need the intercept parameters $[\rho_r]$ to fully determine the cumulative probabilities, in practice the parameter of interest is β_m ” (Agresti, 2010: 48).

overall goodness of fit of our model to the data...However, those who have been concerned about such biases, in conjunction with others, have been surprised by the sense in which the quality of the estimates is affected by various sources of imperfection (of measurement and/or analysis). Thus, by applying the double least squares method, Huang and Sudhir (op. cit.: 339) found the unexpected prevalence of an attenuation bias: “...the OLS approach significantly underestimates the causal effect of satisfaction on loyalty intent [willingness to recommend], suggesting that the combination of attenuation bias as a result of measurement error in customer satisfaction²⁶ and (the unsigned) omitted variable bias overwhelms the inflationary bias caused by the common-methods problem.”

Conclusions

Table 10 summarises the main points of the methodology we developed and proved to be applicable. It draws a parallel between its weaknesses and its strengths. Table 11 outlines various ways in which our contribution could be extended.

Model sophistication: Point 1 of Table 10

Pro. We have shown how to construct robust econometric relationships, rigorously respecting the ordinal categorical nature of the metrics at stake, between, on the one hand, predictors and, on the other, a manifest response underpinned by a latent attitude. As Little (2004: 1855) pointed out: “*Robustness prevents inconsistency and guards against absurdity*”, but this assurance requires a “*a structure that inherently constrains answers to a meaningful range of values.*” (Little, 1970: B470). Such an imperative of credibility has led to remarkable developments, in particular: the so-called attraction models²⁷, such that the market shares extrapolated from them can be neither negative nor greater than 100% and sum to 100%.

Cons. Although realistic, these specifications are unfortunately not elementary. The same applies to the cumulative multinomial model, which was specifically designed for ordinal categorical responses and which we adapted to rigorously estimate the effects of similar antecedents. It may seem as convoluted as those that dominate the literature on brand rivalry and intra product-line cannibalisation (Bultez et al., 1995). For this reason, we have endeavoured to explain and illustrate it in the same way as Brusco (2022) did to popularise the more straightforward binomial-logistic model. For those who wish to master the

²⁶ They do not mention the measurement error of loyalty intention, even though it is only operationalised by a single item.

²⁷ Or *competitive interactions* (Cooper and Nakanishi, 1988; Bultez, 1997), *discrete choice* (Cortsjens and Gautchi, 1983; Aurier and Mejía, 2014), *gravitational* in geomarketing (Cliquet, 1988) and *asymmetric interdependencies* in category management (Bultez et al., 1989).

econometric approach, we have provided a tutorial: **CATORDREG.xlsx** → [hyperlink](#). We have also ensured that it is necessary to use such non-linear formalisations, which will eventually become standard practice²⁸. Many others stick to strictly linear models that treat predictors and responses, albeit qualitative, as quantitative variables. Our approach allows us to test whether they can indeed be treated as such.

Parameterisation and assessment of estimates: points 2 and 3 of Table 10

Pros. The need for multiple regression tools that treat ordinal categorical variables as rigorously as nominal and quantitative variables, whether dependent or predictive, has been demonstrated by the case studied. The approach proposed here contributes to the development and adoption of such tools. It works efficiently by combining optimisation and large-scale resampling simulation. The four times²⁹ 10,000 non-linear programs for estimating the many parameters (nearly twenty) of our model, on large samples, were all solved in a limited number of iterations. The bootstrapping procedure used makes it possible to

- gauge the heterogeneity of perceptions of the scores' meaning, simply by the ranges of the distributions of estimators of their quantified values;
- assess the actual accuracy of the estimates of the satisfaction criteria weights: the magnitude of the standard errors is a function of the imprecision in the quantification of the metrics.

Taking these uncertainties into account, we tested the properties of both the *response* and its *predictors*. At the same time, we showed how to rigorously assess the nature of their measurement scales. Empirically, we have spotted the **NPS** unreliability³⁰.

Although only applied to model the relationship between *satisfaction* and *willingness to recommend*, the methodology used can be adapted to test many other predictors and responses. The literature reviewed suggests other cases where the implementation of our method would be appropriate: the study of the determinants of *repurchase intention* (Bayol et al, 2000; Larivière et al, 2016), the *likelihood to visit a point of sale or an online site* (Mead et al, 2020), *sentiment spontaneously expressed on online review platforms* (Chakraborty et al, 2022), *trust* in a product or service rating system (e.g. nutrition score : Bollinger et al, 2022;

²⁸ A few pioneers have already paved the way for even more complex methodological advances. For example, to assess the impact of product category price variations on the price image of supermarkets, Lourenço et al. (2015: 204) estimated a *cumulative probit multinomial Bayesian model* explaining an ordinal categorical metric to assess the overall level of prices charged by outlets (scoring from 1 = *excellent* to 9 = *very bad*). However, to our knowledge, no one has ever empirically tested the need for such elaborations as we have done here.

²⁹ Two clienteles × two models.

³⁰ Kristensen and Eskildsen (2011) attempted to test it but contented themselves with a questionable graphical clustering. Bultez et al (2024) examined its relevance for benchmarking.

stars in Michelin Guide; power rating of household appliances) or in comparative tests (*Consumer Reports*). All we need to do is redefine the components of (1): the attitudinal resultant, R_i , and its antecedents: $s_{i,m}$ (stimuli).

Cons. To our knowledge, no statistical software package offers a ready-to-use procedure for fitting models that incorporate the principles of monotonic and cumulative multinomial regression. We have therefore developed one. Appendix II lists the SAS code we have programmed. The syntax used and our comments on it should make it easier to adapt to other cases, or even to translate it into other languages. Upon request, the first author will answer any related questions.

Limitations and prospects for extension

Any model is inherently reductive, and ours is no exception: in (6.c), the random component, $\tilde{\epsilon}_i$ at least partially embodies its shortcomings. We have assumed that it is logistic, the default option for cumulative multinomial regressions. Other distributions - the most classical: normal and log-log - are feasible, in which case the expression of the selected distribution function should be substituted in $F(\tilde{\epsilon}_i)$, and in the case that one of them proves to be analytically intractable (e.g., normal distribution), a statistical function can numerically integrate its probability density.

Missing variables. This random component may not be sufficient to account for any omission of explanatory variables. This will be the case (discussed under 1 in Table 11) when there are interactions between the main determinants or between these and third moderating factors. Quantifying these various predictors will require differentiating the incrementation-parameters (δ) that calibrate the intervals between the degrees of their respective measurement scales, if these are not common. The effects of their interactions will be reflected by the coefficients of the products of the predictors quantified in this way.

Mismeasurement. The *ICT* products/services evaluated are hardly differentiated, hence customers who are not very satisfied with their operator could, despite their disenchantment, recommend it if they believe that its competitors underperform. Besides the competitive environment, several other factors that may affect the meaning and reliability of consumer judgements should be considered:

- the scope of the key question: in this case, it concerns the supplier's recommendation, so it is global (multi-product), and not specific to each service,

- the possible shortcomings of the methods used to elicit opinions (modes of questionnaire administration, time needed to answer, sequencing of questions, time it takes to answer them, number of categories making up the scales, semantics used to differentiate them, etc.);
- but also, and above all, the respondents' involvement in expressing their feedback, which varies according to the context in which they are questioned.

Consequently, latent constructs are imperfectly reflected in their purported manifestations. Thus, in addition to specification errors, there are measurement errors, which could be modelled by generalizing systems of structural equations, as suggested in point 2 of Table 11.

Decision support. None of the models we reviewed includes decision variables or performance criteria other than consumer feedback metrics. For our efforts to model their feelings and reactions to be worthwhile, bridges must be built both upstream (actions) and downstream (results). Point 3 of Table 11 discusses how to add these missing links. In terms of antecedents, we need to understand how the components of the products-services offered determine their satisfaction - for example, in the case of *ICT*: surfing and down/up/loading speeds, user-friendliness and stability of connections, (mal)functioning of decoders, accessibility to after-sales service and follow-up interventions, network failures, tariffs...As to subsequent behavior, we need to link attitudes and intentions towards the brand to the actual conduct: share of wallet, loyalty/attraction, responsiveness, conversion, cross-selling, upgrading, etc., and these in turn lead to market share, sales and profitability (of e.g., investments in the development and maintenance of technological infrastructures).

A final word

We hope that our article will not only spark concerns about the true nature of marketing metrics and encourage replications of our approach, but also spur advances along the lines outlined in Table 11, and perhaps beyond.

Table 10. Methodological review

Contributions	Costs	Benefits
1. Model combining monotonic and cumulative multinomial regressions.	Complexity of the <i>latent trait theory</i> underlying the formalization of the cumulative probability of <i>graded response</i> as a function of <i>thresholds</i> .	Rigor: respects the ordinal-categorical nature of predictors and response, by parameterizing degrees of feeling expressions.
2. Empirical calibration of the model	Lack of user-friendliness: no standard statistical procedure.	Optimization: maximization of sample likelihood. Adaptability: wide range of possible applications.
3. Assessment of the precision of estimates and test of the possibilities of assimilation of marketing metrics to interval-level measurements.	Long computer time, requiring efficient mathematical programming routines.	Statistical robustness: complete distributions of estimates resulting from model fits to simulated resampling of data.

Table 11. Avenues for future research

Concerns	Issues	Solution ideas	Challenges
1. Interactions between the main determinants and/or with covariates/moderators.	Calibration of predictors on different scales.	(a) Introduce different sets of parameter increments ($\delta_{j s} \rightarrow \kappa_j$), specific to each type of predictor (main vs. moderator), if they are not measured on a common scale, in order to guarantee the monotonicity of their quantification [equation (4), defining: $s_{i,m}$]. (b) Add the product of the interacting factors to their linear combination [equation (6.b), defining: S_i].	(a) Since the number of parameters rises, the sample must be more informative. (b) All combinations of levels of the factors interfering with one another must be considered.
2. Measurement errors, possibly intercorrelated.	Imperfections in survey methods and data collection contexts affect measurement reliability.	Generalization of structural equation models, so as to respect the ordinal categorical nature of manifest variables and latent constructs, as well.	Increased complexity of the specification of the likelihood function specification and, consequently, of its maximization.
3. Managerial operability.	Effects of action levers on the bottom-line.	Track economic indicators that depend on customer feedback metrics (market share, lifetime value, etc.) and investigate causes of variations in these metrics (post-experience reaction; calls to after-sales service; incident reports, etc.).	Enlargement of the model both downstream (profitability) and upstream (investments). Integrative CRM development.

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Appendix I: Properties of the multinomial cumulative model

I.1. Equation (8. b) determining the response probabilities for $r \in \{1, 2, \dots, D - 1\}$, also holds for $r = 0$ and $r = D$, defined by (8. c) and (8. d).

Proof: $\pi_{i,r} = F(\rho_{r+1} - \mathbb{S}_i^*) - F(\rho_r - \mathbb{S}_i^*)$ implies:

- $\pi_{i,0} = F(\rho_1 - \mathbb{S}_i^*) - F(-\infty - \mathbb{S}_i^*) = F(\rho_1 - \mathbb{S}_i^*)$, since $\rho_0 = -\infty$ et $F(-\infty) = 0$;
- $\pi_{i,D} = F(\infty - \mathbb{S}_i^*) - F(\rho_D - \mathbb{S}_i^*) = 1 - F(\rho_D - \mathbb{S}_i^*)$, since $\rho_{D+1} = \infty$ et $F(\infty) = 1$.

I.2. The consistency of the model is also evidenced by the fact that the sum of the response probabilities equals one.

Proof: Extremes aside, each response probability is equal to the difference between two values of the distribution function F , and each first term of this difference is equal to the second term of the higher response probability ($r + 1$). By adding them together, these terms cancel out:

$$\begin{aligned} \sum_{r=0}^{D+1} \pi_{i,r} &= F(\rho_1 - \mathbb{S}_i^*) + [F(\rho_2 - \mathbb{S}_i^*) - F(\rho_1 - \mathbb{S}_i^*)] + [F(\rho_3 - \mathbb{S}_i^*) - F(\rho_2 - \mathbb{S}_i^*)] \\ &\quad + \dots + [F(\rho_{D+1} - \mathbb{S}_i^*) - F(\rho_D - \mathbb{S}_i^*)] + [1 - F(\rho_D - \mathbb{S}_i^*)] = 1. \end{aligned}$$

I.3. If F is the standardized logistic function, then the interval between two latent attitude categorization boundaries is the logarithm of the odds ratio of events: $(\tilde{A}_i < \tau_{r+1})$ and $(\tilde{A}_i < \tau_r)$. That is: $\Delta_{r|R} = \ln(\text{Odds}(\tilde{A}_i < \tau_{r+1}) / \text{Odds}(\tilde{A}_i < \tau_r))$.

Proof: $\text{Odds}(\tilde{A}_i < \tau_d) = P(\tilde{A}_i < \tau_d) / [1 - P(\tilde{A}_i < \tau_d)] = e^{(\rho_d - \mathbb{S}_i^*)}$, therefore the odds ratio grow exponentially with the interval size:

$$\text{Odds}(\tilde{A}_i < \tau_{r+1}) / \text{Odds}(\tilde{A}_i < \tau_r) = e^{\rho_{r+1}} / e^{\rho_r} = e^{(\rho_{r+1} - \rho_r)} = e^{\Delta_{r|R}},$$

size which is thus equal to the Napierian logarithm of this ratio.

Appendix II: Computer programming

Monotonic or *categorical*³¹ regression techniques - used to calibrate linear models³² analogous to (5) - are not suitable for estimating the parameters of nonlinear relationships of the type (8. b, c, d).

³¹ Methods that *qualitative regression* generalizes to "cases where the dependent variable and the independent variables are of any type (nominal, ordinal, or numerical)" and "which consists in coding the qualitative variables, respecting their nominal or ordinal structures, in such a way as to maximize the multiple correlation between the dependent variable and the independent variables" (Tenenhaus, 1979: 5). The monotonic regression method has been revisited by Troutt (1989). It can be carried out using the *TRANSREG* procedure of SAS and the *CATREG* function of SPSS, but these tools do not allow us to impose a common optimal coding for several explanatory variables.

³² A typical example is conjoint analysis, often used, even to design a sales force incentive plan: bonuses related to sales quotas René Darmon, 1979: 137-139).

Table 12. Optimal parameterization routine

I. Estimation of the standard multinomial cumulative model on untransformed variables
$s_{i,m} \rightarrow$ prefix: S, suffixes: IT ($m = 1$), TV ($m = 2$), MB ($m = 3$) et QP ($m = 4$); $R_i \rightarrow$ LikR
PROC LOGISTIC DATA=Binaries DESCENDING OUTEST=Estimates; MODEL LikR = SIT STV SMB SQP / DIST=MULTINOMIAL LINK=CUMLOGIT;
II. Procedure for maximizing the log-likelihood of the sample of observations
PROC OPTMODEL;
II.1. Input of data previously coded as binary variables
Response: recommendation likelihood, coded: $B_{i \rightarrow r} \rightarrow$ prefix: BR_ ; suffixes : 0 à 10. Satisfaction ratings: $b_{i \rightarrow j m} \rightarrow$ prefix : BS ; suffix designating the product-service (m) : IT, TV, MB, QP, followed by degree of satisfaction: _0,_1,_2,_3,_4.
SET Client; NUMBER BR_0{Client}, BR_1{Client}, BR_2{Client}, BR_3{Client}, ... , BR_10{Client}; NUMBER BSIT_0{Client}, BSIT_1{Client}, BSIT_2{Client}, BSIT_3{Client}, BSIT_4{Client}; NUMBER BSTV_0{Client}, BSTV_1{Client}, BSTV_2{Client}, BSTV_3{Client}, BSTV_4{Client}; NUMBER BSMB_0{Client}, BSMB_1{Client}, BSMB_2{Client}, BSMB_3{Client}, BSMB_4{Client}; NUMBER BSQP_0{Client}, BSQP_1{Client}, BSQP_2{Client}, BSQP_3{Client}, BSQP_4{Client}; READ DATA Binaries into Client=[_n_] BR_0 BR_1 BR_2 BR_3 ... BR_10 BSIT_0 BSIT_1 BSIT_2 BSIT_3 BSIT_4 BSTV_0 BSTV_1 BSTV_2 BSTV_3 BSTV_4 BSMB_0 BSMB_1 BSMB_2 BSMB_3 BSMB_4 BSQP_0 BSQP_1 BSQP_2 BSQP_3 BSQP_4;
II.2. Initialization of parameter values using estimates from step I
Increments to be squared: $\nabla_{j S} \rightarrow$ prefix: d_S ; $\nabla_{r R} \rightarrow$ prefix: d_R. First response threshold: $\rho_1 \rightarrow$ Rho_1. Weights of satisfaction criteria: $\beta_m \rightarrow$ prefix: Beta_S
VAR d_S{1..4} init 1, d_R{1..9} init 1, Rho_1 init 1.609, Beta_SIT init 0.5, Beta_STV init 0.4, Beta_SMB init 0.6, Beta_SQP init 0.7;
II.3.1. Determination of response thresholds, obtained by cumulating squared increments: $\nabla_{r/R}^2$
$\rho_r, r \in \{2, 3, \dots, 10\} \rightarrow$ prefix: Rho_
IMPVAR Rho_2 = Rho_1+(d_R[1]*d_R[1]); IMPVAR Rho_3 = Rho_2+(d_R[2]*d_R[2]); IMPVAR Rho_9 = Rho_8+(d_R[8]*d_R[8]); IMPVAR Rho_10 = Rho_9+(d_R[9]*d_R[9]);
II.3.2. Quantification of satisfaction ratings:
Scale levels: $K_{j S} \rightarrow$ prefix: K_ ; $\nabla_{j S} \rightarrow$ prefix: d_S, $s_j \rightarrow$ prefix: Q_S. Application of the adjustment factor: G/K_G , for: $G = 4$.
IMPVAR K_1 = (d_S[1]*d_S[1]); IMPVAR K_2 = K_1+(d_S[2]*d_S[2]); IMPVAR K_3 = K_2+(d_S[3]*d_S[3]); IMPVAR K_4 = K_3+(d_S[4]*d_S[4]); IMPVAR Q_SIT {i in Client} = (K_1*BSIT_1[i] + K_2*BSIT_2[i] + K_3*BSIT_3[i] + K_4*BSIT_4[i])*(4/K_4); IMPVAR Q_STV {i in Client} = (K_1*BSTV_1[i] + K_2*BSTV_2[i] + K_3*BSTV_3[i] + K_4*BSTV_4[i])*(4/K_4); IMPVAR Q_SMB {i in Client} = (K_1*BSMB_1[i] + K_2*BSMB_2[i] + K_3*BSMB_3[i] + K_4*BSMB_4[i])*(4/K_4); IMPVAR Q_SQP {i in Client} = (K_1*BSQP_1[i] + K_2*BSQP_2[i] + K_3*BSQP_3[i] + K_4*BSQP_4[i])*(4/K_4);
II.4. Definition of overall satisfaction: $S_i^* = \sum_{m=1}^M \beta_m \cdot s_{i,m} \rightarrow$ GS
IMPVAR GS {i in Client} = Beta_SIT*Q_SIT[i] + Beta_STV*Q_STV[i] + Beta_SMB*Q_SMB[i] + Beta_SQP*Q_SQP[i];
II.5.1. Evaluation of the logistic probability distribution of responses:
$\Lambda(\rho_r - \sum_{m=1}^M \beta_m \cdot s_{i,m}) \rightarrow$ Prefix : LAMBDA. $\Lambda(x) = 1/(1 + e^{-x})$
IMPVAR LAMBDA1 {i in Client} = 1/(1+EXP(-(Rho_1-GS[i]))); IMPVAR LAMBDA10 {i in Client} = 1/(1+EXP(-(Rho_10-GS[i])));
II.5.2. Calculation of response probabilities: $\pi_{i,r} \rightarrow$ prefix: Pi
IMPVAR Pi0 {i in Client} = LAMBDA1[i]; IMPVAR Pi1 {i in Client} = LAMBDA2[i] - LAMBDA1[i]; IMPVAR Pi2 {i in Client} = LAMBDA3[i] - LAMBDA2[i]; IMPVAR Pi8 {i in Client} = LAMBDA9[i] - LAMBDA8[i]; IMPVAR Pi9 {i in Client} = LAMBDA10[i] - LAMBDA9[i]; IMPVAR Pi10 {i in Client} = 1 - LAMBDA10[i];
II.5. Maximization of the log-likelihood of the observations: $\mathcal{L} \rightarrow$ LogLikALL, formula: (10)
IMPVAR LogProbClient {i in Client} = log(Pi0[i])*BR_0[i] + log(Pi1[i])*BR_1[i] + log(Pi2[i])*BR_2[i] + ... + ... + log(Pi9[i])*BR_9[i] + log(Pi10[i])*BR_10[i]; MAX LogLikALL = SUM{i in Client} LogProbClient[i]; SOLVE;

*Cumulative multinomial regression*³³ quantifies only the response, whereas in (9) $\mathbb{S}_i^*$ - as defined in (8. a) - requires that these M antecedents also be quantified. Furthermore, these predictors, which result from judgments of the same type - multiple expressions of satisfaction on the same scale, from the same subjects - must be coded identically. Therefore, we solved problem (10) using a general mathematical optimization tool, the:

OPTMODEL procedure in **SAS**. Table 12 details and comments the code generated to fit our model to the data related to the *ICT*-case discussed in the **Implementation** section.

Estimation. Given the risk of algorithmic convergence to a local optimum (Tenenhaus, 1979: 14, 16, 18 and 21) or divergence (an even higher risk), the parameters were initialized with caution: we start the iterations of our maximization heuristic with the values obtained from the results of the application of the standard cumulative logistic regression [cf. Step *I*] on the $s_{i,m}$ and not on the $\mathbb{S}_{i,m}$, which amounts to imposing the *interval* scale: $\delta_{j|s} = 1, \forall j$.

Inference. Any empirical deduction about parameters based on the usual statistics (standard deviations, Student's t , p -values) would be risky, since these standards do not take into account the fact that the G hidden parameters - the intervals between the levels of the satisfaction scale ($\delta_{j|s}$) - are also estimated from the same data set. However, the use of bootstrapping (Chernick, 1999) makes it possible

- to avoid underestimating the imprecision of the $\hat{\beta}$ et $\hat{\rho}_r$, consequence of this ignorance,
- and, more importantly, to assess the precision of the $\hat{\kappa}_{j|s}$ and thus test whether the ordinal categorical ratings could have been interpreted quantitatively.

This bootstrapping method - sometimes called naive or primitive - consists of repeating the analysis on numerous samples generated by *random (uniform or weighted) and with replacement* resampling of the available observations, as described in Table 13. According to Efron and Tibshirani (1994: 15), bootstrapping allows “*the data analyst...to use any estimator, no matter how complicated, with the assurance that he or she will also have a reasonable idea of the estimator's accuracy*”. These two authorities (1986: 62) believe that 50 to 100 bootstrapped samples suffice for estimating standard deviations, but that many more - more than 1,000 - are needed for confidence intervals. To ensure the reliability of our inferences, we simulated many more in each case: 10,000!

The complete SAS code incorporating the procedure commented on in Table 12, run tens of thousands of times on as many bootstrapped samples, is available on request from the first author.

³³ **PROC LOGISTIC (LINK=CUMLOGIT)** de **SAS**.

Table 13. Inference algorithm based on analyses of samples selected from the parent sample

I.	Set the number of so-called bootstrapped samples: $\mathcal{B}$, sufficiently large.
II.	From $b = 1$ to $b = \mathcal{B}$, (1) Select from the initial parent sample: $E = \{R_i, s_{i,1}, \dots, s_{i,M} \mid i = 1, \dots, N\}$, randomly³⁴ and with replacement, a sample³⁵, denoted: $E^{(b)}$ of the same size, of combinations of the observations on the $M + 1$ variables³⁶ : $E^{(b)} = \{R_u^{(b)}, s_{u,1}^{(b)}, \dots, s_{u,M}^{(b)} \mid u = 1, \dots, N\};$ (2) Re-estimate the parameters through maximization of the log-likelihood of the bootstrapped sample, $\mathcal{L}^{(b)}$, from $E^{(b)}$: $\mathcal{L}^{(b)} = \left[\sum_{i=1}^{i=N} \sum_{r=0}^{r=10} B_{i \rightarrow r}^{(b)} \cdot \ln \pi_{i,r}^{(b)} \right],$ using the <i>PROC OPTMODEL</i> specified in Table 12 (part: II); (3) Records estimation results: $\hat{\Theta}^{(b)} := \{ \text{les } \hat{\beta}_m^{(b)}, \hat{\rho}_r^{(b)}, \hat{\kappa}_{j S}^{(b)} \}.$
III.	Infer the properties of the estimators of all parameters, through descriptive syntheses of the distributions of the $\mathcal{B}$ sets of estimation results $\hat{\Theta}^{(1)}, \dots, \hat{\Theta}^{(\mathcal{B})}$: histograms, boxplots, means: $\hat{\theta}$, standard deviations: $\hat{\sigma}(\hat{\theta})$ and precisions: $z(\hat{\theta})$: $\hat{\theta} = \sum_{b=1}^{b=\mathcal{B}} \hat{\theta}^{(b)} / \mathcal{B}, \quad \hat{\sigma}(\hat{\theta}) = \sqrt{\sum_{b=1}^{b=\mathcal{B}} [\hat{\theta}^{(b)} - \hat{\theta}]^2 / (\mathcal{B} - 1)}, \quad z(\hat{\theta}) = \frac{\bar{\theta}}{\hat{\sigma}(\hat{\theta})}.$

³⁴ Either uniformly, or according to a probability distribution determined by population strata representativeness quotas.

³⁵ In the *bootstrapping* literature, it is customary to distinguish bootstrapped samples by the upper index: (b) .

³⁶ For example, by applying the SAS *SURVEYSELECT* procedure, method: *PPS_WR*, for weighted random sampling.

Appendix III: Statistical Indicators

Graphical Representations. Simulation results are presented in the form of histograms and boxplots. Boxplots are connected by the means (indicated by the symbol: +) of the distributions that these graphs summarize. The rectangle circumscribes the central 50% by the limits of the interquartile range: distance between the upper and lower sides of the rectangle. The median lies where the horizontal bar appears inside each rectangle. The whiskers bound the normal range of estimates, which consists of the values within the invisible fences that are located at 1.5 times the mid-spread, down from the first quartile (25% centile) and up from the third quartile (75% centile) respectively. Estimates that fall outside these fences are therefore considered to be abnormally far from the mass of others; these outliers are indicated by squares.

Goodness of fit. The gamma coefficient is defined by:

$$\Gamma = (\mathcal{C} - \mathcal{D})/(\mathcal{C} + \mathcal{D}),$$

where: $\mathcal{C}$ (versus $\mathcal{D}$) is the number of pairs of observations of responses concordant (versus discordant) with their predictions. If $\mathcal{C} = 0$, then $\Gamma = -1$, while if $\mathcal{D} = 0$, then $\Gamma = +1$. Applying Γ to the cumulative logistic regression requires us to rely on the predicted response for each subject, in our case:

$$\hat{R}_i = \sum_{r=0}^{r=10} r \cdot \hat{\pi}_{i,r}.$$

Considering two individuals: i and λ , with different responses: $c \neq d$, so that:

$$B_{i \rightarrow c} = 1 \text{ and } B_{\lambda \rightarrow d} = 1,$$

predictions for them are judged to be

- concordant if $c \geq d \Rightarrow \hat{R}_i \geq \hat{R}_\lambda$,
- and discordant when $c \geq d \Rightarrow \hat{R}_i \leq \hat{R}_\lambda$.